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ORIGINAL ARTICLE

Constraining spectral induced polarization-derived pore size distributions in carbonates using MIP and SEM

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Summary
Characterizing pore size distribution (PSD) in carbonate reservoirs remains a significant challenge due to their complex and heterogeneous pore structures. This study introduces a novel integration of spectral induced polarization (SIP) with advanced image analysis and conventional petrophysical techniques to non-destructively quantify PSD and infer flow-related properties. A key innovation lies in the application of a modified open-source MATLAB algorithm to invert SIP spectra and extract relaxation time distributions, which were then modeled using a Double Cole–Cole function. The SIP-derived PSDs showed strong agreement with mercury intrusion porosimetry (MIP) data, validating the approach. Furthermore, high-resolution scanning electron microscopy (SEM) images were processed using a watershed-based segmentation technique to resolve microporosity and complement SIP estimates. While both MIP and SEM are destructive methods, they were used solely as benchmarking tools to constrain and validate SIP interpretations. A quantitative relationship between SIP quadrature conductivity and critical petrophysical parameters such as grain size variation and permeability were evaluated using existing theoretical models. The study demonstrates the potential of SIP as a standalone predictive tool for reservoir quality assessment, reducing reliance on destructive laboratory methods. These findings contribute new insights into the electro facies behavior of carbonates and offer a scalable approach for subsurface characterization in complex reservoir settings.

Keywords Complex conductivity · Grain size distribution (GSD) · SIP- relaxation time distribution (RTD) · Image processing · Rock physics · Double Cole-Cole model

Introduction

Carbonate rocks exhibit heterogeneity in their textures and properties, resulting in a varied distribution of flow characteristics. These rocks are significant for the storage of oil, gas, and water and account for more than 60% of global oil production and 40% of natural gas production (Kargarpour 2020; Weiqiang et al., 2020). Carbonate reservoirs are also host to some of the largest freshwater storage systems (Cunningham et al., 2012; Kovács 2021). Due to diagenetic processes referring to the physical and chemical changes after deposition—these rocks display mineralogical, structural, and textural heterogeneity (Lu et al., 2021) at various scales, from micrometers to kilometers (Brown 2011). Despite their importance, there remains a significant gap in the literature regarding accurate characterization of PSD at the laboratory scale in carbonate systems and its implications for fluid flow. Carbonate porosity has been classified based on depositional

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origin (Choquette and Pray, 1970), petrophysical fabric (Lucia 1983, 1995), and hybrid schemes (Lonoy 2006a, b). Under these highly variable conditions, realistic estimation of petrophysical properties becomes complex and uncertain (Bust et al., 2011). Numerous laboratory methods have been developed to estimate PSD. Among these, MIP (Cnudde et al. 2009; Sun et al. 2020; Bohnsack et al., 2021) and nuclear magnetic resonance (NMR) (Zheng et al., 2020; Hussein et al., 2018; Niu and Zhang 2018; Zhang et al., 2018b) are considered industry standards. However, MIP is destructive and may alter fine pore structures, while NMR NMR requires time-consuming sample preparation and its interpretation using certain models can overestimate larger pores if homogeneity in pore geometry is assumed (Stingaciu et al. 2010 In NMR-based pore size distribution analysis, the use of a fixed or arbitrary T_2 cutoff to distinguish bound and free water can introduce uncertainties. Different strategies exist to estimate fluid mobility, and an inappropriate cutoff may misclassify pore fluids, leading to biased estimates of permeability and storage capacity (Coates et al., 1999). SEM is often used to visualize pore structures at high resolution and extract 2D pore metrics while the imaging itself is non-destructive, the sample preparation process is typically destructive and limited in spatial coverage (Diamond and Leeman 1994; Liu et al., 2020; Safari et al., 2021a, b). To improve accuracy, some studies have employed integrated approaches particularly in siliciclastic settings (Revil et al., 2012; Cao et al., 2016; Zhang et al., 2018a, 2022) integrated μ-CT, MIP, NMR, and SIP to characterize sandstone pore structures and recommended using MIP as a reference for aligning pore size distributions. Their approach focused on resolution-based comparison across methods. Recent work by August et al. (2025), building on the earlier study by Garcia et al. (2018), in the Santos Basin (Brazil) demonstrated that integrating NMR and resistivity data can significantly improve estimates of pore-throat size and permeability in carbonate formations, outperforming traditional NMR-only approaches. Their calibration-free workflow highlighted the limitations of relying on arbitrary T_2 cutoffs and emphasized the value of combining electrical and petrophysical measurements. Inspired by such efforts, this study extends the concept by integrating SIP data constrained using MIP and SEM to offer a more physically grounded and non-destructive method for characterizing PSD in heterogeneous carbonate systems (Yao et al., 2014). Existing SIP studies on carbonates often assume a simplified pore system. Some focus on calcite precipitation or saturation effects under controlled conditions, without addressing the full complexity of natural carbonate rocks. While useful for understanding specific processes, these studies do not capture pore-scale heterogeneity (Garcia et al., 2024). To overcome these limitations, this study integrates SIP data with independent measurements from MIP and SEM pore-scale heterogeneity of carbonates. This approach ensures that the interpretation of complex conductivity measurements is directly supported by physical evidence from the rock's pore structure, enhancing the reliability of SIP-derived PSD estimation in carbonate systems. Carbonate rocks present additional challenges because of their complex and multi-modal pore structures, including micropores, intercrystalline pores, and vugs, which produce overlapping polarization signals. Previous studies have demonstrated that such heterogeneity can complicate SIP interpretation. For instance, Müller-Huber et al. (2018) used the Cole-Cole model in grain-dominated and mud-dominated carbonate rocks to derive the Cole-Cole parameters (c, τ, m) and compared the outcomes with NMR and MIP data. They observed that grainstones showed higher Cole-Cole exponent (c) values, while mudstones had lower values, indicating textural control on polarization behavior. Zhang et al. (2018b) estimated PSD using a joint inversion of SIP and NMR for an oomoldic dolostone sample, but observed significant mismatches between the two datasets, highlighting challenges in SIP interpretation without proper constraints. While many laboratory techniques exist to estimate PSD, their integration with SIP remains largely unexplored in carbonate rocks. This study aims to fill this gap by investigating the use of SIP to characterize PSD using the Double Cole-Cole model, and validating the results using complementary techniques. A modified open-source MATLAB code is used to invert SIP spectra and extract RTDs. SEM images are processed with a watershed algorithm to extract detailed pore body information, and MIP data serve as quantitative benchmarks.

Although both MIP and SEM are destructive, they are used here solely for validation. To our knowledge, this is the first study that systematically constrains SIP-derived PSD estimates in carbonate rocks using both image-based and porosimetry-based techniques. The results not only improve understanding of the relationship between electrical polarization and pore architecture, but also highlight the potential of SIP as a standalone, non-destructive tool for reservoir quality assessment in complex carbonate systems.

Theory

PSD in carbonates can be quantified from electrical polarization responses by interpreting SIP data through complex conductivity models. In SIP, the response results from ionic polarization at the mineral–fluid interface, particularly within the electrical double layer (EDL). At low frequencies (< 100 Hz), both EDL and membrane polarization can significantly influence the measured response, while at higher frequencies (> 1 kHz), mechanisms such as Maxwell-Wagner

and dielectric polarization may dominate (Revil et al., 2017; Li et al., 2021). If the EDL is the primary contributor to polarization, then the RTD derived from SIP spectra can be interpreted as a proxy for PSD (Tong et al., 2006; Revil and Florsch, 2010; Revil et al., 2014; Jougnot et al., 2010).

The method resolves both conduction and polarization processes by decomposing the complex conductivity into in-phase and quadrature components, the latter being closely related to pore-scale capacitive effects.

SIP and relaxation phenomena

SIP, or complex conductivity, characterizes the combined conductive and dielectric properties of porous media across a broad frequency range (0.1–45 kHz). The measured complex conductivity (σ^*_ω) comprises an in-phase component (σ′) and a quadrature component σ″, which are expressed as follows (Halisch et al., 2014; Osterman et al., 2016):

$$\sigma' = \frac{1}{F}(\sigma_w + \sigma'_s) \tag{1}$$

$$\sigma'' = \sigma''_s \tag{2}$$

Here, σ_w denotes the pore water conductivity, σ_s' and σ_s'' represent the real and imaginary parts of the surface conductivity, respectively, and F is the formation factor.

At low frequencies, polarization arises primarily from the electrical double layer (EDL) at the mineral-fluid interface. The characteristic relaxation time τ_0 represents the time required for ions in the electrical double layer (EDL) to return to their equilibrium positions after the applied electric field is removed (Vinegar 1988, 1984; Schwarz 1962; Revil et al., 2012; Liu et al., 2014; Coperey et al., 2019).

$$\tau_0 = \frac{r^2}{aD_i} \tag{3}$$

where r is the pore radius, D_i is the ionic diffusion coefficient, and a is a geometric scaling factor. This relationship forms the basis for inferring pore-scale features from relaxation spectra.

Forward model and inversion scheme

The inversion framework employed in this study is based on the generalized linear admittance approach developed by Florsch et al. (2014), which incorporates a time-varying generalized least squares (TVGLS) scheme Tarantola and Valette (1982). The model interprets SIP spectra as a superposition of Debye-type relaxation mechanisms, represented through a continuous distribution of relaxation times. The forward response is expressed as:

$$\sigma^* = \sigma_\infty + \frac{\sigma_0 - \sigma_\infty}{1 + i\omega\tau_0} + i\omega C \tag{4}$$

where σ_∞ is the high-frequency conductivity, g(τ) is the normalized relaxation time distribution (RTD), and C accounts for high-frequency capacitive contributions.

. Equation (19) represents the ideal Debye-type relaxation, where frequency dispersion is not explicitly included (i.e., c = 1). To generalize this formulation, the model can be expressed as:

$$\sigma^* = a_R - \int_0^\infty g_\tau(\tau)\,\phi(\omega,\tau)\,d\tau + i\omega C \tag{5}$$

where $g(\tau)$ is the normalized RTD and $\phi(\omega,\tau) = 1/(1+i\omega\tau)$ is the kernel function. A logarithmic change of variables ($\tau = e^s$, $\omega = e^{-z}$) enhances numerical stability and ensures consistent treatment across the experimental frequency range of 0.1 mHz to 45 kHz. Here, Consistent treatment refers to numerically stable and uniform processing of relaxation time distributions for all measured frequencies across the experimental range, avoiding bias or accuracy loss between low and high frequencies. The resulting discretized system is inverted using the TVGLS scheme to obtain a stable RTD with appropriate regularization (see Appendix A for details).

Double Cole–Cole representation

While the original model incorporates a single relaxation domain as described by the classic Cole–Cole formulation, the heterogeneity of carbonate systems often induces multi-scale polarization behavior. To accommodate this, the model was extended to include a Double Cole–Cole expression (Pelton et al., 1978; Schwartz and Furman 2012; Tartrat et al., 2019):

$$\sigma^*_\omega = \sigma_\infty\left[1 - \frac{M_1}{1 + i\omega\tau_1{}^{c1}} - \frac{M_2}{1 + i\omega\tau_2{}^{c2}}\right] \tag{6}$$

where M_1 and M_2 are chargeability terms, τ_1 and τ_2 are characteristic relaxation times, and c_1 and c_2 are Cole–Cole exponents corresponding to two distinct relaxation domains. The angular frequency is given by $\omega = 2\pi f$. The indices 1 and 2 represent low and high frequency dispersions, respectively.

This dual-component model improves the ability to resolve overlapping polarization features that are common in grain-supported and matrix-supported domains within

carbonates. Analytical expressions for the real and imaginary components of Eq. (6) are presented in Appendix A.

Mercury intrusion porosimetry (MIP)

MIP is a widely used technique for estimating the pore-throat size distribution of porous materials by quantifying the volume of mercury intruded into the sample under controlled pressure increments. The method is based on the principle that non-wetting fluids, such as mercury, require external pressure to enter pore throats, with smaller pores requiring higher intrusion pressures. The relationship between the applied pressure and the corresponding pore diameter is governed by the Washburn equation (Washburn 1921; Abell et al., 1999):

$$P = \frac{-4\gamma\, cos\theta}{d} \tag{7}$$

Where, P = Pressure, ϒ = Surface tension of the liquid, θ = Contact angle of the liquid and d = diameter of the capillary.

Despite its destructive nature, MIP remains a standard method due to its capability to resolve a broad range of pore throat sizes. In this work, MIP results are used to validate and constrain the SIP-derived relaxation time distributions.

Scanning electron microscopy (SEM)

Scanning Electron Microscopy (SEM) provides high-resolution images that enable qualitative and quantitative assessment of pore morphology, size distribution, and connectivity in carbonate rocks. SEM complements spectroscopic and porosimetry methods by visually resolving pore features that span sub-micron to millimeter scales. While SEM does not directly yield volumetric pore size distribution like MIP or SIP, it allows for structural interpretation and morphological classification of pore networks, which can inform and validate PSD trends derived from other techniques (Zhang et al., 2015; Widiatmoko et al., 2010; Safari et al., 2021a, b; Salvini et al., 2023).

Material and methodology

Eleven carbonate rock samples were collected from the Tertiary carbonate platforms located offshore of Western India, corresponding to stratigraphic units from the Middle Eocene to Early Oligocene. These samples were selected as representative of platform carbonates and encompass a range of depositional and diagenetic textures, rendering them suitable for detailed multi-scale pore structure characterization.

Petrographic observations and petrophysical analyses revealed a complex pore architecture dominated by intercrystalline porosity typically resulting from dolomitization alongside vuggy and channel-type porosities formed through secondary dissolution processes. Vuggy porosity is characterized by isolated or interconnected cavities produced by selective dissolution, whereas channel porosity consists of irregular, elongated voids that often cross-cut the primary fabric, commonly associated with enlarged fractures or stylolite seams (Irajian et al., 2016; Sharma et al., 2013).

The measured porosity ranged from 0.07 to 0.22 (fraction), and permeability varied from 0.00496 mD to 10.94 mD. Porosity was determined using the conventional dry and saturated weight method, while permeability was measured using a steady-state gas permeameter and calculated according to Darcy's law This wide range of petrophysical properties reflects the intrinsic heterogeneity of the carbonate matrix and establishes a robust foundation for evaluating the sensitivity of spectral induced polarization (SIP) to pore-scale variations in geometry and connectivity.

SIP data acquisition

SIP measurements were conducted on 11 carbonate core samples using the ZEL-SIP04-V02 impedance meter (Zimmermann et al., 2008), across a frequency range of 0.1 Hz to 45 kHz. The samples were obtained from well-log cores drilled in Tertiary carbonate formations offshore western India. Core lengths varied between 3.8 cm and 6 cm, with an average diameter of approximately 2.5 cm. Prior to measurement, all samples were oven-dried at 60 °C for 48 h to eliminate residual moisture. They were then saturated under vacuum with tap water for one month to ensure complete saturation and minimize the presence of entrapped air, which can distort polarization responses.

The saturating fluid (tap water initially) had an electrical conductivity of approximately 0.079 S/m. For subsequent experiments, additional salinities were introduced. The second conductivity was 0.58 S m^{-1}, after one month of equilibration with the samples. For the third salinity condition, the pore water conductivity was 5 S m^{-1}. Under the highest salinity condition, the pore water conductivity was 9.18 S m^{-1}. The three salinity concentrations corresponding to these solutions were 2.6 g L^{-1}, 37 g L^{-1}, and 73.1 g L^{-1}, respectively.

A four-electrode configuration was employed, using non-polarizable Ag/AgCl electrodes, with two electrodes for current injection and two for voltage measurement as shown in

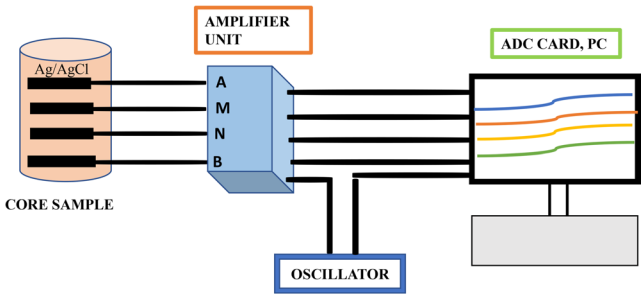


Fig. 1 Experimental setup to conduct SIP measurements using ZEL impedance meter. The sample is connected to the ZEL impedance meter with current electrodes (A & B) and potential electrodes (M & N)

Fig. 1. This arrangement reduces electrode polarization effects and ensures accurate acquisition of the complex impedance spectrum. The measured complex impedance Z_{ω}^{*} was converted to complex conductivity σ_{ω}^{*} using the relation:

$$\sigma'' = -\frac{L}{A}\frac{(Imag\,(R))}{Abs\,(R)^2} \tag{8}$$

$$\sigma' = \frac{L}{A}\frac{(Real\,(R))}{Abs\,(R)^2} \tag{9}$$

where L is the sample length, A is the cross-sectional area, and Real (R) and Imag (R) is the frequency-dependent impedance, Abs(R) is the absolute value of impedance.

All measurements were conducted at ambient laboratory conditions (~25 °C). Both single and double Cole–Cole models were used to extract relaxation parameters for subsequent analysis.

Mercury intrusion porosimetry (MIP) data acquisition

Mercury intrusion porosimetry (MIP) was carried out using the PoreMaster 60 (Quanta Chrome Instruments) to determine the pore throat size distribution of the carbonate samples. Approximately 2 g of powdered sample was placed into the chamber for analysis. The instrument operates in two pressure regimes: an initial low-pressure stage (up to 50 psia) used for sample evacuation, mercury filling, and macropore detection, followed by a high-pressure stage reaching up to 60,000 psia for the intrusion of progressively finer pores.

Pore throat diameters were calculated automatically by the instrument software based on the Washburn equation, which relates applied pressure to capillary diameter. These calculations rely on standard values for surface tension and contact angle predefined within the instrument settings, consistent with established mercury properties and measurement protocols.

Scanning electron microscopy (SEM) data acquisition

A Carl Zeiss EVO 18 scanning electron microscope (SEM) was used for image acquisition. The instrument provides a magnification range from < 5× to 1,000,000× and a resolution of up to 3072 × 2304 pixels. For sample preparation, SEM stubs were first cleaned with ethylene glycol and then mounted using carbon tape. The coated samples were imaged using SEM, and scale bars of 1 µm, 2 µm, 10 µm, 20 µm, 100 µm, and 200 µm, were applied to capture and calibrate features across various pore scales.

Pore throat size distribution Estimation from SEM

To quantify pore throat dimensions, SEM images were processed through a sequential image analysis workflow. First, a median filter was applied to suppress high-frequency noise and enhance structural clarity. The images were then converted into binary form using Otsu's thresholding, which separated the pore space from the solid matrix. Subsequently, the watershed algorithm was employed to delineate individual pore throats. A distance transform was then applied to the binary image to generate a topographic surface, where pore centers corresponded to local minima. The watershed transform was used to simulate the flooding process from these minima, allowing the separation of adjacent pores by placing watershed lines at the narrowest constrictions. These dividing lines correspond to pore throats, allowing adjacent pores to be clearly separated. This approach enabled the identification of constrictions within the pore network, serving as proxies for pore throat diameters (Gonzalez and Woods 2018).

Grain size distribution Estimation from SEM

Grain size distribution was estimated using ImageJ software (Schneider et al., 2012, Schindelin et al., 2012). SEM images were first calibrated by assigning scale to pixel dimensions. Grain boundaries were manually delineated, and the longest axis of each grain was measured. A statistically significant number of grains were analyzed per image to ensure representativeness. The resulting measurements were compiled to construct grain size distribution histograms for comparative analysis.

Results & discussion

Evaluation of Cole–Cole model for SIP inversion

To analyze the relaxation behavior of carbonates, we employed an open-source MATLAB code (Florsch et al.,2014) to fit the experimental SIP data with empirical models such as Cole-Cole and Double Cole-Cole. Initially, the aim was to assess the suitability of the Cole-Cole model for capturing the frequency-dependent complex conductivity response. As illustrated in Fig. 2(a) and (b), the Cole–Cole model effectively matched the SIP spectra of homogeneous texture samples (e.g., I14 and I15); however, it failed to adequately represent the relaxation time distribution (RTD), as shown in Fig. 2(c). In contrast, for samples with heterogeneous textures (e.g., I25 and I41), the model did not provide satisfactory fits to either the in-phase or quadrature components of the data.

This discrepancy underscores the model's limitations when applied to carbonates exhibiting complex pore structures. SEM image analysis further supported this interpretation. As shown in Figure 3, homogeneous samples display uniformly distributed intercrystalline porosity, while heterogeneous samples exhibit dual-porosity systems characterized by both fracture and intracrystalline porosity. These microstructural variations strongly influence relaxation behavior and require more flexible models for accurate interpretation.

Although Figs. 2 and 3 highlight a subset of representative samples (I14, I15, I25, and I41) to demonstrate contrasting textures, the entire suite of 11 samples was analyzed using the same SIP–MIP–SEM workflow. The results were consistent across the full dataset, as reflected in Table 1.

Double Cole–Cole inversion and Cross-Validation

To better represent the multi-scale relaxation processes in carbonates, the Double Cole–Cole model was implemented. Figure 4 shows the improved fitting of SIP spectra using this model across representative samples. The estimated model

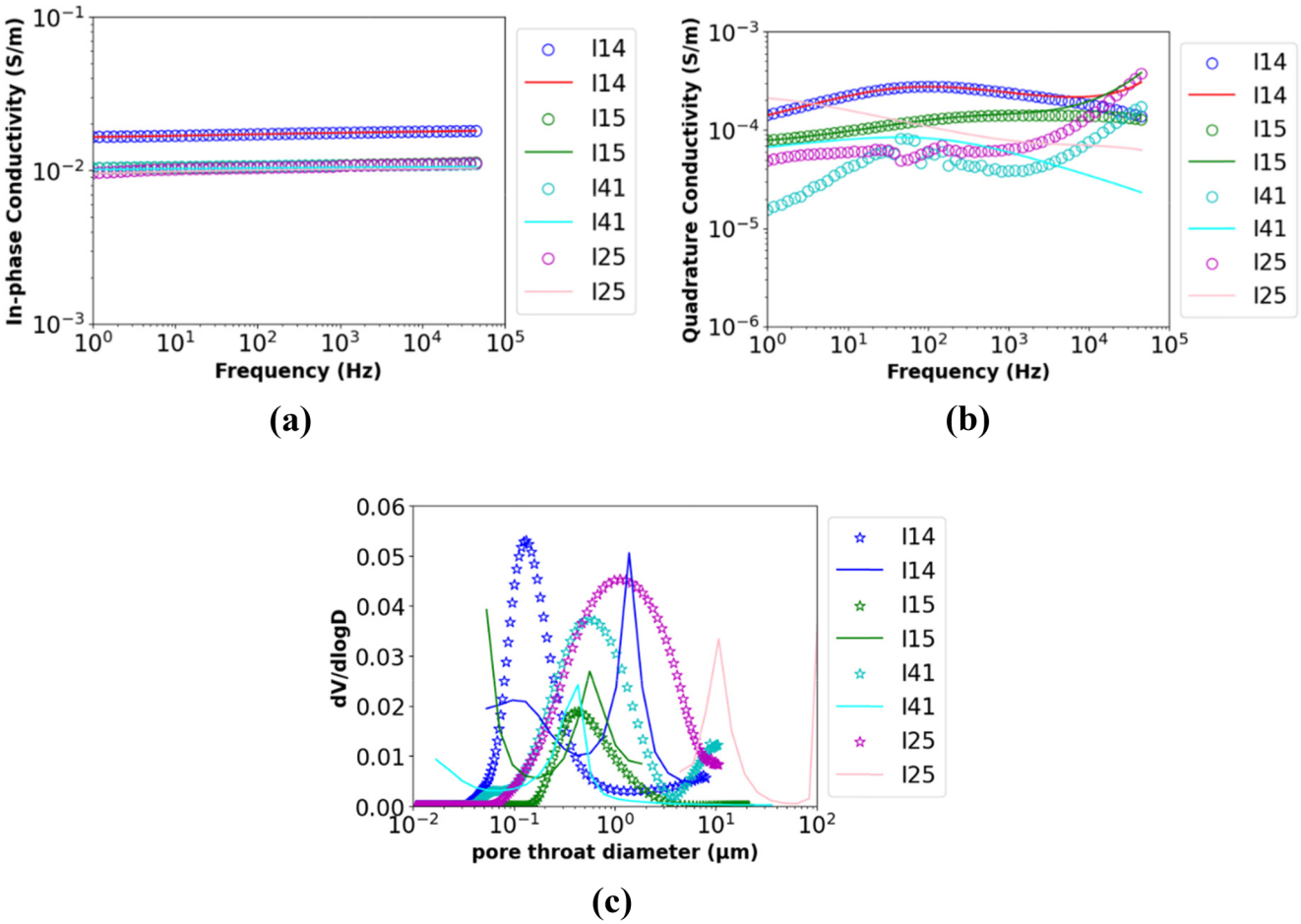


Fig. 2 **a** & **b**: SIP spectra (**a**) In-phase and (**b**) Quadrature conductivity, of carbonate samples fitted with Cole-Cole model. **c** Pore size distribution of carbonate samples fitted with inverted SIP data. (*) represents the experimentally measured pore size distribution using MIP, while the line illustrates the estimated dominant pore size distribution obtained through analysis with the RTD code

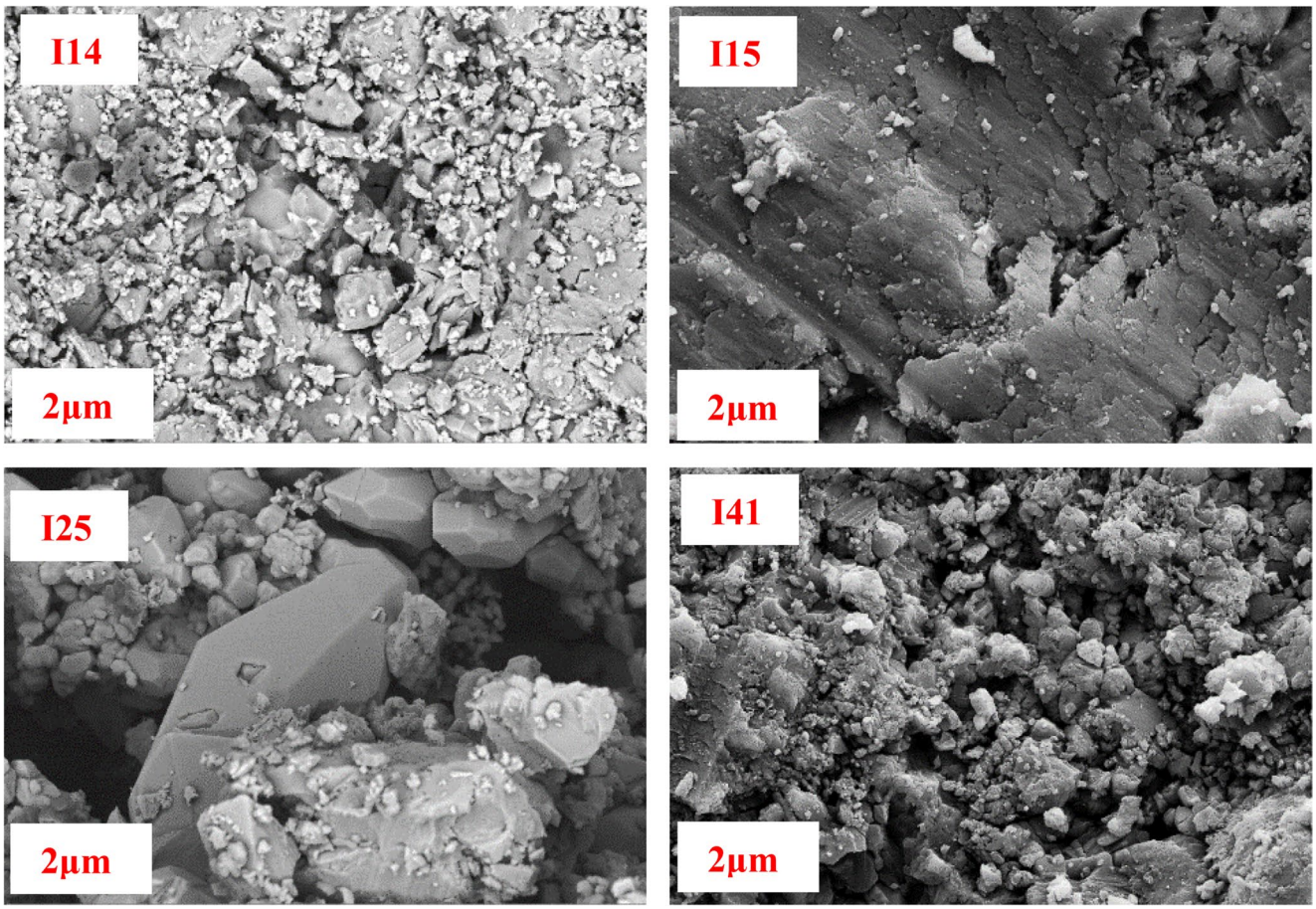


Fig. 3 Back Scattered (BSE) SEM backscattered images of carbonate sample

Table 1 Inverted double Cole-Cole parameters used to fit the laboratory complex conductivity data. The inversion is done using the RTD code as described in appendix A

Sample ID	σ_∞	M_1	M_2	c_1	c_2	τ_1	τ_2	Real/Quad error
I14	0.0184	0.0025	0.0008	0.45	0.2	3.16	0.0015	4.17E-05/1.04E-05
I18	0.073	0.0044	0.00046	0.6	0.4	10	0.0155	2E-04/9.9E-06
I22	0.0128	0.000275	0.00198	0.6	0.3	10	1E-07	1.59E-05/2.74E-06
I25	0.011	0.00010	0.0020	0.81	0.1	2.56	0.01	5E-04/1.16E-05
I8	0.0095	0.00014	0.00148	0.7	0.3	2.25	0.1	4.21E-05/1.11E-05
I15	0.0095	0.00043	0.0050	0.45	0.1	10	2E-04	5.33E-05/4.75E-06
I48	0.0036	8.86E-06	8.51E-05	0.8	0.18	17.01	0.017	1.17E-05/1.76E-06
I4	0.0074	0.00013	0.00092	0.8	0.2	2.21	0.1	7.04E-05/4.44E-06
I41	0.010	0.00090	0.00422	0.7	0.3	10	0.0028	1.43E-04/1.09E-05
I20	0.10	0.0135	0.0125	0.7	0.6	11.42	0.1	8.0E04/1.0E-04
I21	0.0059	0.000149	0.00079	0.9	0.1	3.59	1E-10	7.24E-05/1.39E-06

parameters, including M_1, M_2, τ_1, τ_2, c_1, c_2, and $\sigma\infty$, are listed in Table 1, along with the root-mean-square errors for real and quadrature spectra.

The inversion was performed using a generalized least-squares approach, and the RTDs were obtained using the kernel-based framework outlined in Appendix A. Compared to the single-pole Cole–Cole model, the Double Cole–Cole formulation successfully captured the broader and often bimodal RTDs observed in these rocks. The two relaxation domains identified by the double Cole–Cole model are interpreted as arising from distinct textural controls within the sample as shown in Fig. 5, with shorter and longer relaxation times reflecting differences in the matrix structure. While the fitted parameters indicate the presence

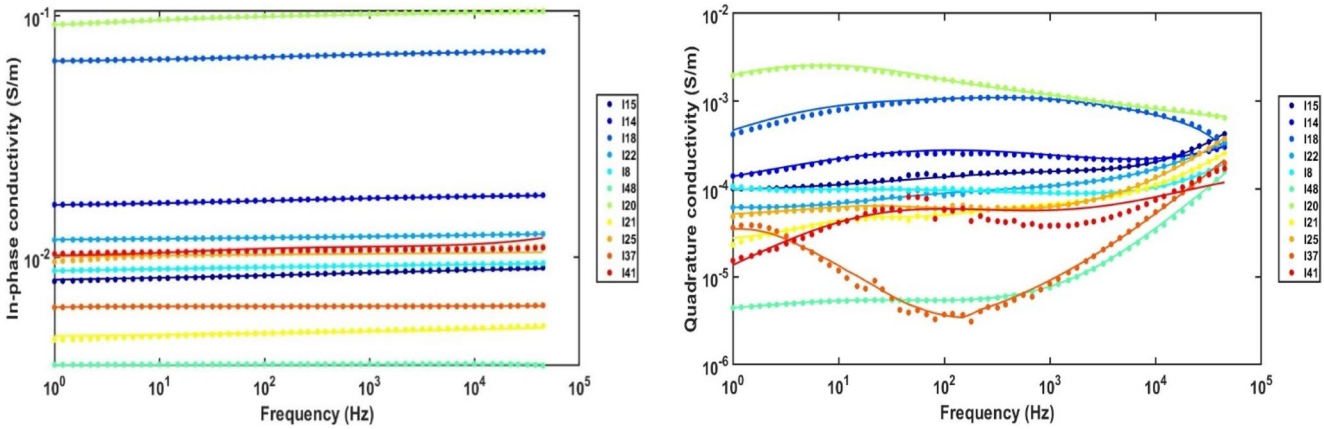


Fig. 4 SIP spectra fitted with Double Cole-Cole model. Circles displays measured SIP data, while Line depicts the fitted data

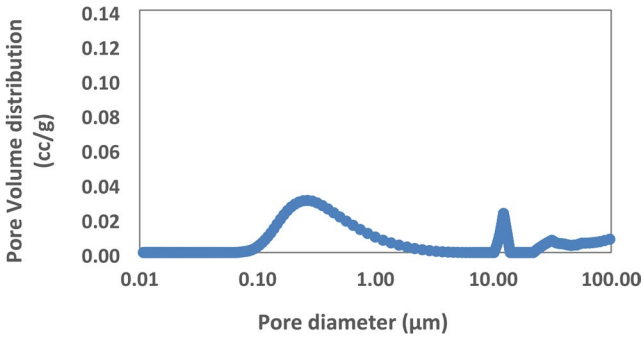


Fig. 5 Multiple dispersion observed in one of the carbonate samples from our dataset

of two separate polarization mechanisms, this interpretation is presently qualitative and requires further quantitative validation. Inversions were stable across multiple initial guesses and noise levels, suggesting the solutions are robust even without formal confidence intervals. Future work may benefit from applying formal uncertainty quantification techniques, such as Monte Carlo or Bayesian inversion, to derive statistical confidence bounds.

Figure 6 presents the inverted PSDs obtained from SIP alongside those derived from MIP measurements. A strong agreement was observed between the two methods, with similar peak positions and distribution shapes

Table 2 displays the differences in the dominant peak and standard deviation of pore volume between the SIP and MIP measurements. It can be seen that the values of pore volume distributions estimated using inversion of SIP are very close to the values measured with MIP, showing the accuracy of RTD code for the carbonate samples.

The better fit and relaxation time provided by the Double Cole-Cole model for our carbonate samples suggest that it might capture some underlying complexities not adequately described by a single relaxation process. Carbonates can exhibit complex dielectric behavior due to various factors, including mineralogy, porosity, fluid content, and structural characteristics (Markov et al., 2012; Gomaa et al., 2015).

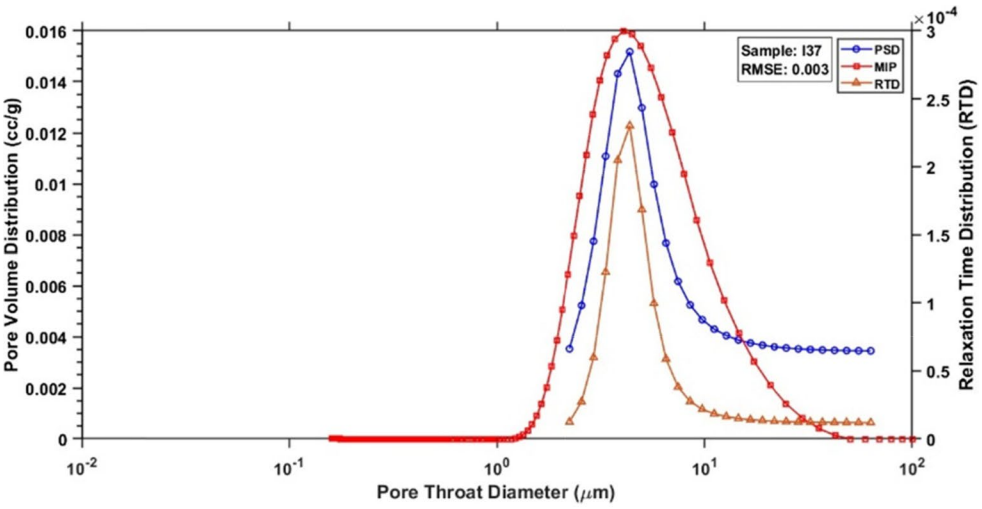
SEM-Based validation of pore throat distributions

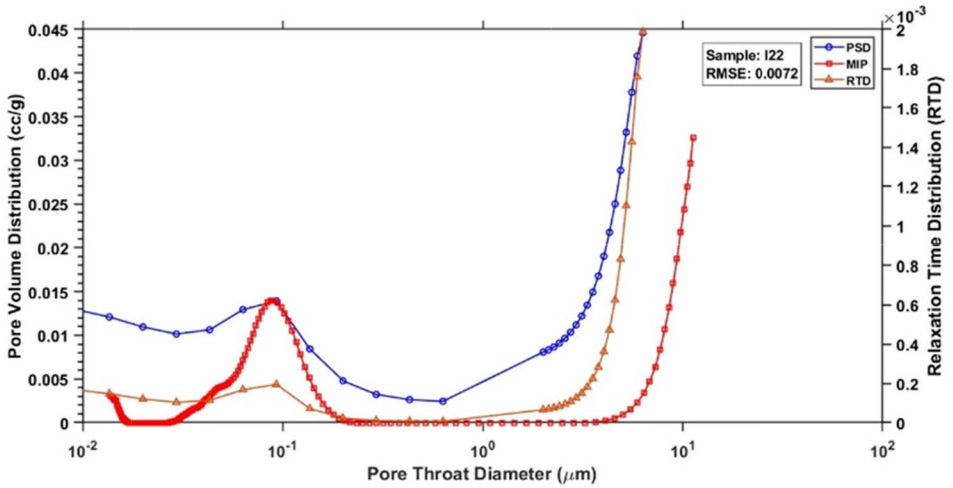
SIP-derived PSDs were further validated using 2D SEM image analysis. Pore throat sizes were extracted through image processing workflows applied to high-resolution SEM images captured at multiple magnifications. As shown in Figs. 7(i)–(v), the dominant pore throat diameters fell within the 0.09–50 μm range. This finding is consistent with the PSDs estimated via both SIP and MIP, confirming the reliability of the SIP inversion framework. According to Lonoy (2006a, b), pores within this size range are classified as microporous, which is significant for understanding carbonate storage and flow behavior.

While SEM analysis was initially limited to a subset of samples for image clarity and resolution management, the workflow can be readily extended to all other samples. The agreement across SIP, MIP, and SEM methods confirms the presence of microporosity in the studied carbonate formations and supports the applicability of the developed workflow to broader carbonate datasets.

Permeability model

To extend the analysis beyond relaxation behavior, we explored the potential of spectral induced polarization (SIP) parameters to estimate permeability in carbonate rocks. Initially, 11 carbonate core samples were investigated. To improve statistical robustness and better capture the variability in lithological textures and petrophysical properties, the dataset was extended to include approximately 30 additional carbonate samples, comprising wackestone, mudstone, and limestone lithologies from the Tertiary carbonate platform offshore of Western India (Fig. 8).





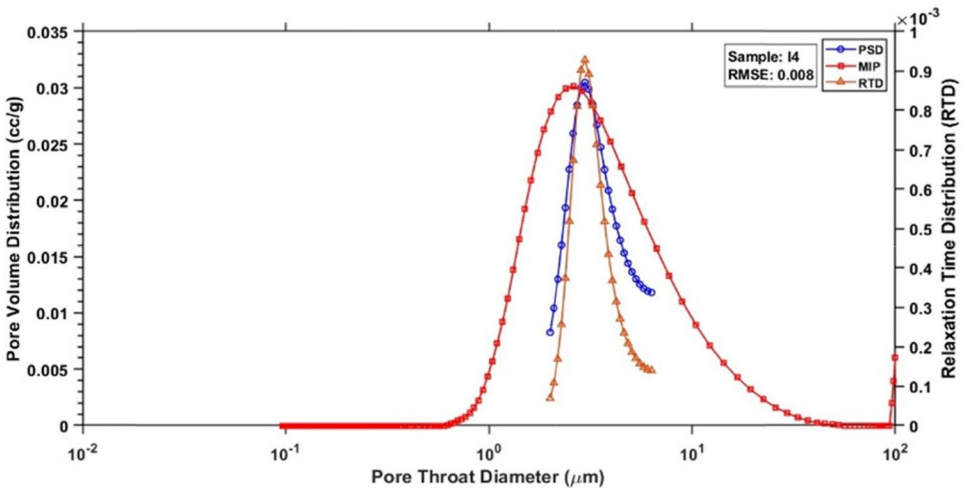


Fig. 6 **a** Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD, RTD is triangle red line, retrieved from RTD code (Florsch et al. 2014). The values of standard deviation, median and peak of the pore volume are given in Table 2. **b** Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD, RTD is triangle red line, retrieved from RTD code (Florsch et al., 2014). The values of standard deviation, median and peak of the pore volume are given in Table 2. **c** Comparison of PSD from MIP (Dotted line) and from inversion of SIP data (blue line) using RTD, RTD is triangle red line, retrieved from RTD code (Florsch et al., 2014). The values of standard deviation, median and peak of the pore volume are given in Table 2

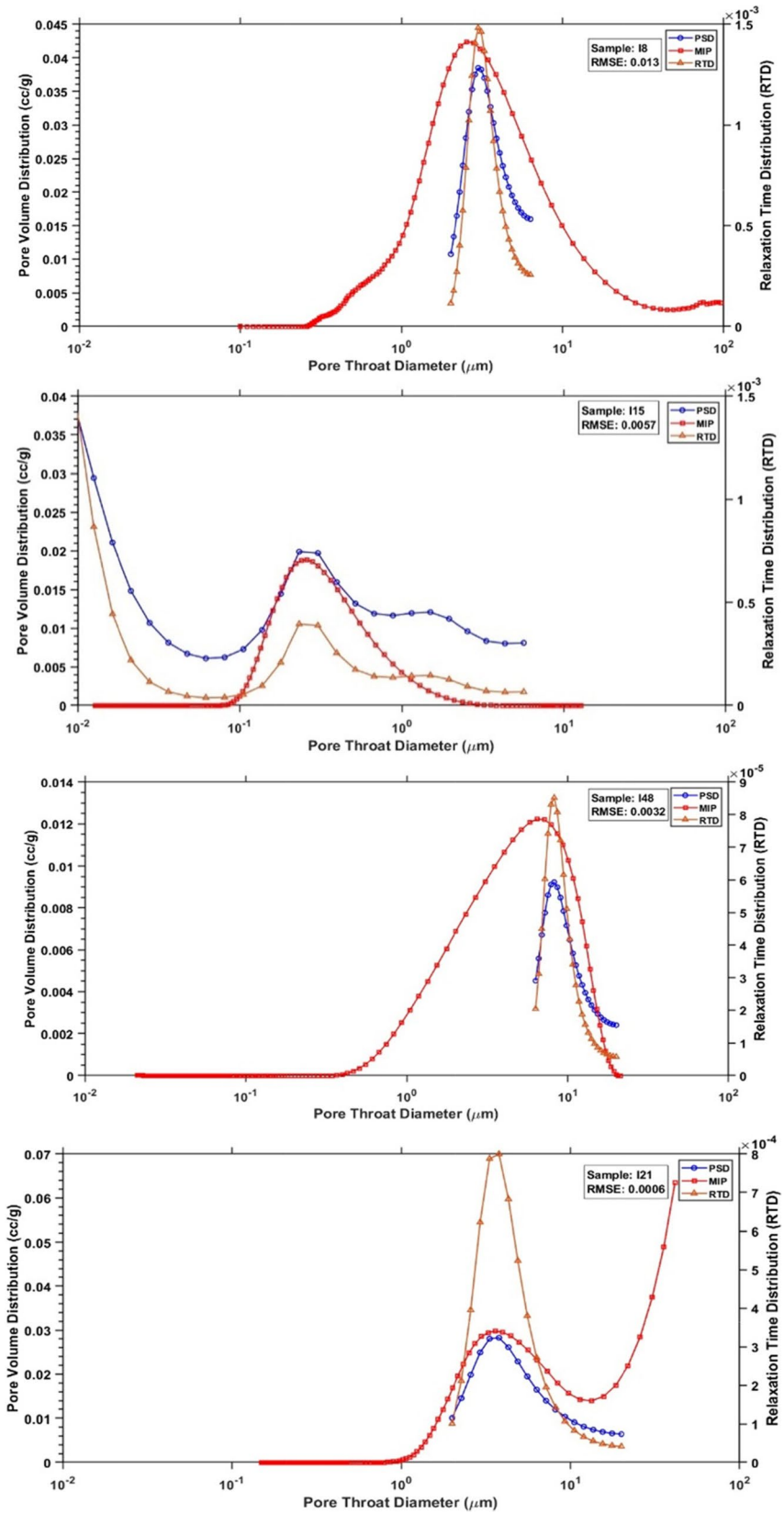


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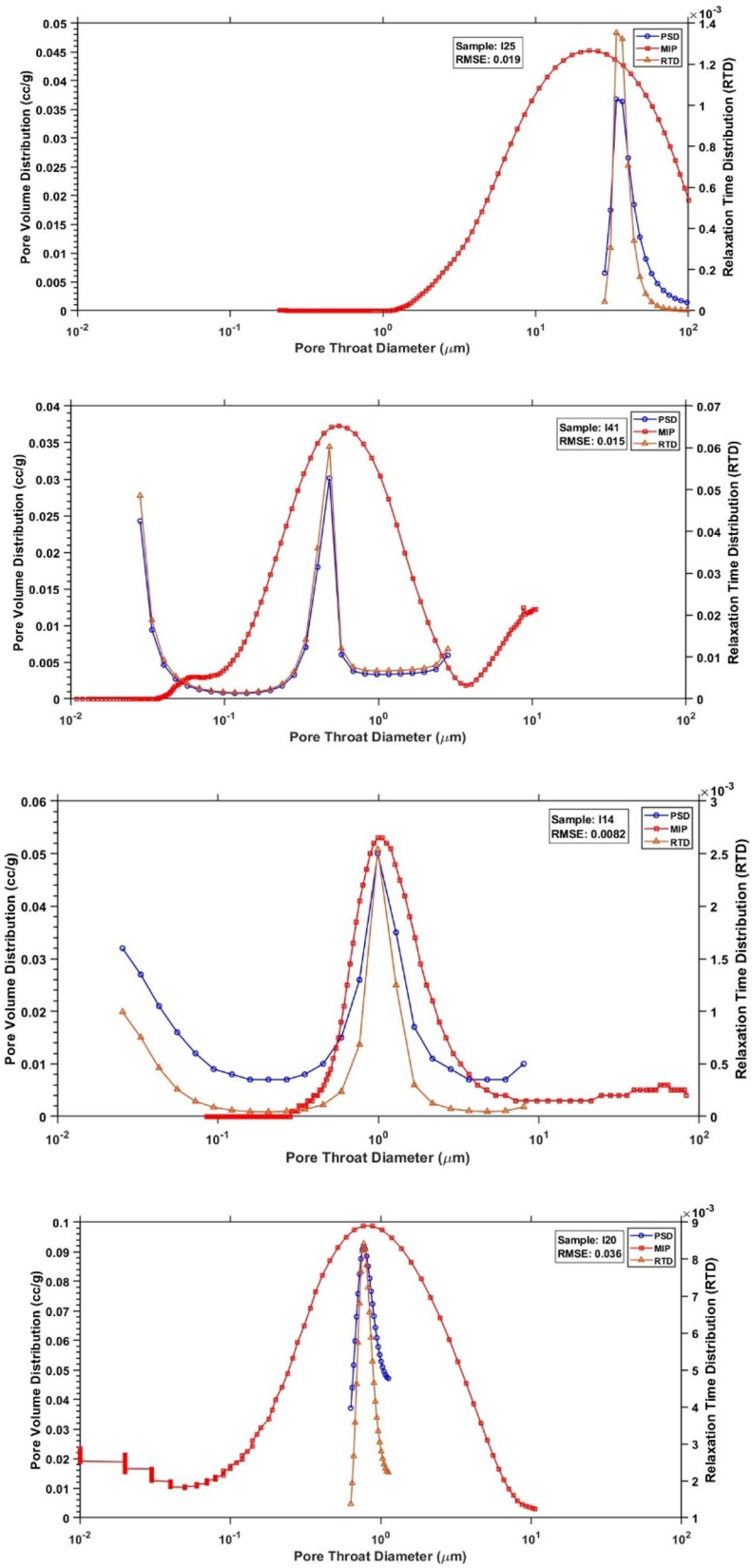


Fig. 6 (continued)

Table 2 Estimated values of standard deviation, and peak of the pore volume distribution of MIP and SIP data, where 'r_c (peak) µm' is the peak value of the pore volume distribution

Sample ID	Standard deviation (MIP data)	Standard deviation (SIP data)	r_c(peak) µm (MIP data)	r_c (peak) µm (SIP data)
I14	0.013	0.011	0.053	0.050
I18	0.012	0.0158	0.06	0.07
I22	0.0037	0.0048	0.0139	0.0137
I25	0.0129	0.010	0.05	0.04
I8	0.0106	0.0085	0.04	0.038
I15	0.0046	0.0039	0.018	0.019
I48	0.0031	0.0024	0.01	0.009
I4	0.0074	0.0069	0.03	0.03
I41	0.0092	0.0072	0.04	0.03
I20	0.021	0.017	0.10	0.09
I21	0.0082	0.0077	0.03	0.028

The permeability of these samples was estimated using three SIP-based predictive formulations originally developed for sandstones namely, those proposed by Revil and Florsch (2010), Weller et al. (2015a), and Pape et al. (1987). Because the microstructural and surface conduction characteristics of carbonate rocks differ significantly from those of sandstones, each model was modified and calibrated to better represent the carbonate dataset. The model coefficients were empirically optimized based on measured permeability data. Also, these models were chosen because they relate permeability to formation factor through different physical mechanisms, allowing comprehensive evaluation of how quadrature conductivity incorporation improves prediction accuracy across various theoretical foundations commonly applied in carbonate reservoirs.

The Revil and Florsch (2010) model relates permeability (k) to quadrature conductivity (σ″), formation factor (F), and the specific surface conductivity of the Stern layer (Σs), as follows:

$$k \approx \frac{(\sum s)^2 \sigma''^{-2}}{4.5F^3} \tag{10}$$

A standard value of $\Sigma s = 2 \times 10^{-9}$S was adopted, consistent with Revil and Florsch (2010). The formation factor (F) was determined from the relationship between in-phase conductivity and pore-water conductivity in log–log space using a nonlinear inversion procedure. The best-fit curve provides estimates of F, m, and surface conductivity (σ_s), following the approach described in Panwar et al. (2021). In the original formulation, m = 1.5 was used, corresponding to spherical grain packing. Quadrature conductivity at 1 Hz was selected because it lies centrally in the SIP spectrum, providing a stable, representative value widely used in literature to ensure consistent comparison across samples and studies (Börner & Schön, 1991; Weller et al., 2010b). For the present carbonate dataset, which is more compacted, m = 2.1 was applied to better represent the pore geometry.

The model proposed by Revil and Florsch (2010) demonstrated the best agreement with the observed data. It is important to note, however, that Revil and Florsch (2010) highlighted a limitation of their formulation that it is not applicable to rocks characterized by a continuous solid phase where the Stern layer is absent. In particular, limestones typically do not exhibit a pronounced Stern layer, meaning that the model's assumptions may not strictly hold for all carbonate lithologies. In the present study, the carbonate samples analyzed display sufficient surface-exposed porosity and textural heterogeneity, which enable Stern layer polarization to occur.

The modified version yielded permeability predictions in closer agreement with experimental observations, as shown in Fig. 8, and outperformed other empirical formulations.

$$k \approx \frac{(\sum s)^2 \sigma''^{-2}}{8.82F^3} \tag{11}$$

Permeability for the carbonate core samples was measured using a steady-state gas permeameter under confining pressure. The SIP-derived estimates were calibrated against these measurements. Using the original formulation, the comparison yielded a correlation of R^2 = 0.72, while the modified version with m=2.1 improved the correlation to R^2 = 0.80. This demonstrates the dependency of permeability predictions on the choice of m and highlights the.

improved suitability of the modified Revil model for carbonate rocks exhibiting diverse textures and porosities.

In addition to the Revil formulation, the Weller et al. (2015a) model was examined, which for sandstone was expressed as:

$$K = \frac{2.66 * 10^{-7}}{\sigma''^{0.66} F^{5.35}} \tag{12}$$

This empirical formulation captures the dependence of permeability on SIP-derived quadrature conductivity (σ″) and the electrical formation factor (F). To adapt it for carbonates, the model coefficients and exponents were optimized using nonlinear regression in log–log space. The modified form, calibrated against measured carbonate permeability, is expressed as:

$$K = \frac{7.14 * 10^{-13}}{\sigma''^{0.45} F^{2.39}} \tag{13}$$

The optimized constants yielded an R^2 = 0.573 (log-space fit), indicating a moderate correlation between measured and predicted permeability. This suggests that while

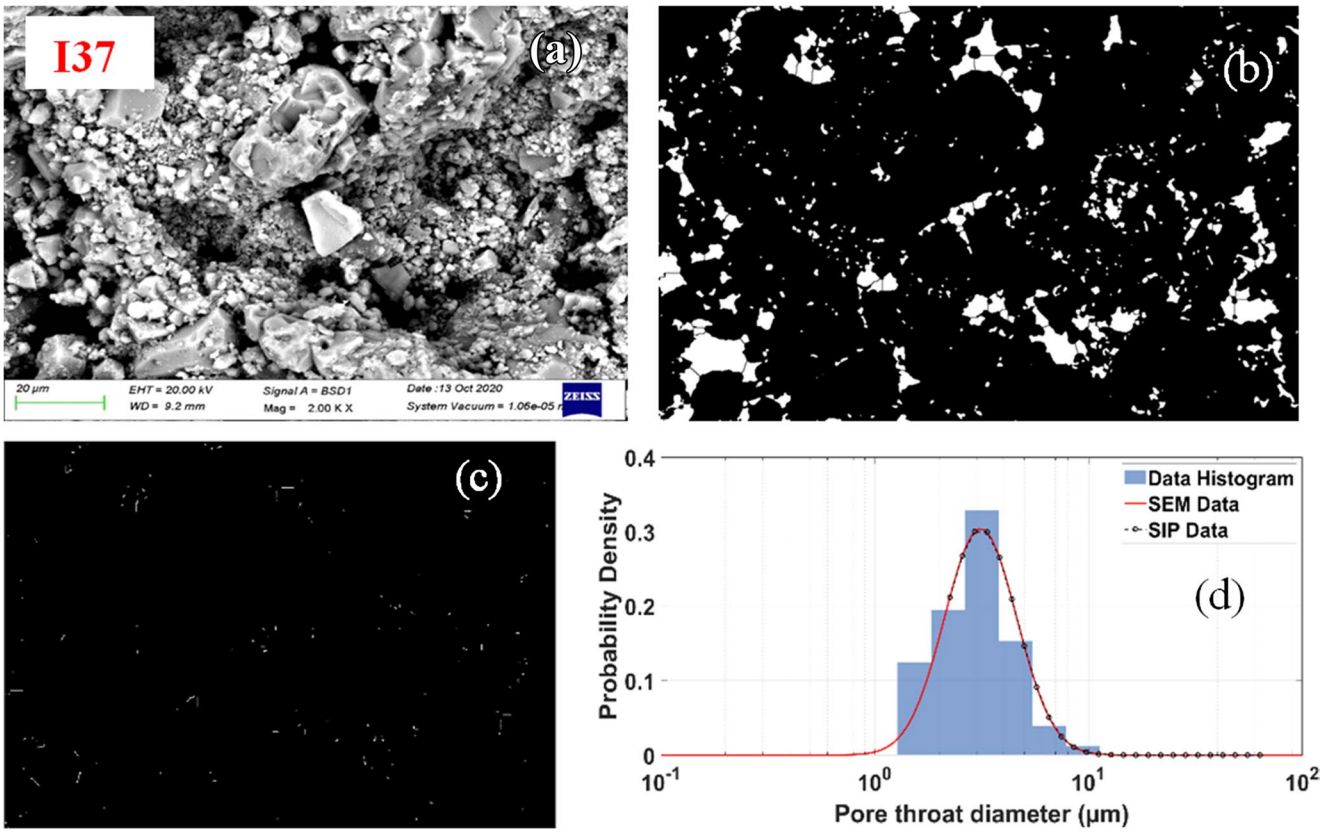


Fig. 7 (**i**): **a** & **b** showing BSE SEM image (20 µm) and segmented image, respectively. **c** The White space indicates the pores while the black area represents the grains in segmented image. **d** Pore throat size distribution of sample I37 using SEM image analysis and SIP. The blue color is showing the histogram, red line is its normal distribution and black dotted line with circles is showing the SIP distribution. (**ii**): **a** & **b** showing BSE SEM image (10 µm) and segmented image, respectively. **c** The White space indicates the pores while the black area represents the grains in segmented image. **d** Pore throat size distribution of sample I14 using SEM image analysis and SIP. The blue color is showing the histogram, red line is its normal distribution and black dotted line with circles is showing the SIP distribution. (**iii**): **a** & **b** showing BSE SEM image (10 µm) and segmented image, respectively. **c** The White space indicates the pores while the black area represents the grains in segmented image. **d** Pore throat size distribution of sample I20 using SEM image analysis and SIP. The blue color is showing the histogram, red line is its normal distribution and black dotted line with circles is showing the SIP distribution. (**iv**): **a** & **b** showing BSE SEM image (100 µm) and segmented image, respectively (**c**) The White space indicates the pores while the black area represents the grains in segmented image. **d** Pore throat size distribution of sample I21 using SEM image analysis and SIP. The blue color is showing the histogram, red line is its normal distribution and black dotted line with circles is showing the SIP distribution. (**v**): **a** & **b** showing BSE SEM image (2 µm) and segmented image, respectively (**c**) The White space indicates the pores while the black area represents the grains in segmented image. **d** Pore throat size distribution of sample I41 using SEM image analysis and SIP. The blue color is showing the histogram, red line is its normal distribution and black dotted line with circles is showing the SIP distribution

the modified Weller model captures general permeability trends, its empirical structure originally tuned for sandstone pore geometries may not fully represent the complex micro-textural variability of carbonates.

Similarly, the Pape et al. (1987) model, initially derived for sandstones, expressed permeability as:

$$K = \frac{475 * 10^{-12}}{Spor^{3.1} F} \tag{14}$$

Where *S*por represents the specific surface area. Since carbonate textures exhibit greater heterogeneity and broader pore-size distributions than sandstones, this model was recalibrated for the carbonate dataset through regression-based optimization:

$$K = \frac{1.35 * 10^{-10}}{Spor^{0.82} F^{1.54}} \tag{15}$$

The modified Paris model achieved an R^2=0.72, indicating a substantial improvement compared to its original sandstone calibration and showing good predictive capability for the carbonate dataset. Among the three formulations, the modified Revil model provides the most robust correlation (R^2 = 0.80), followed by the Paris (R^2 = 0.72) and Weller ($R^2 \approx 0.6$) models.

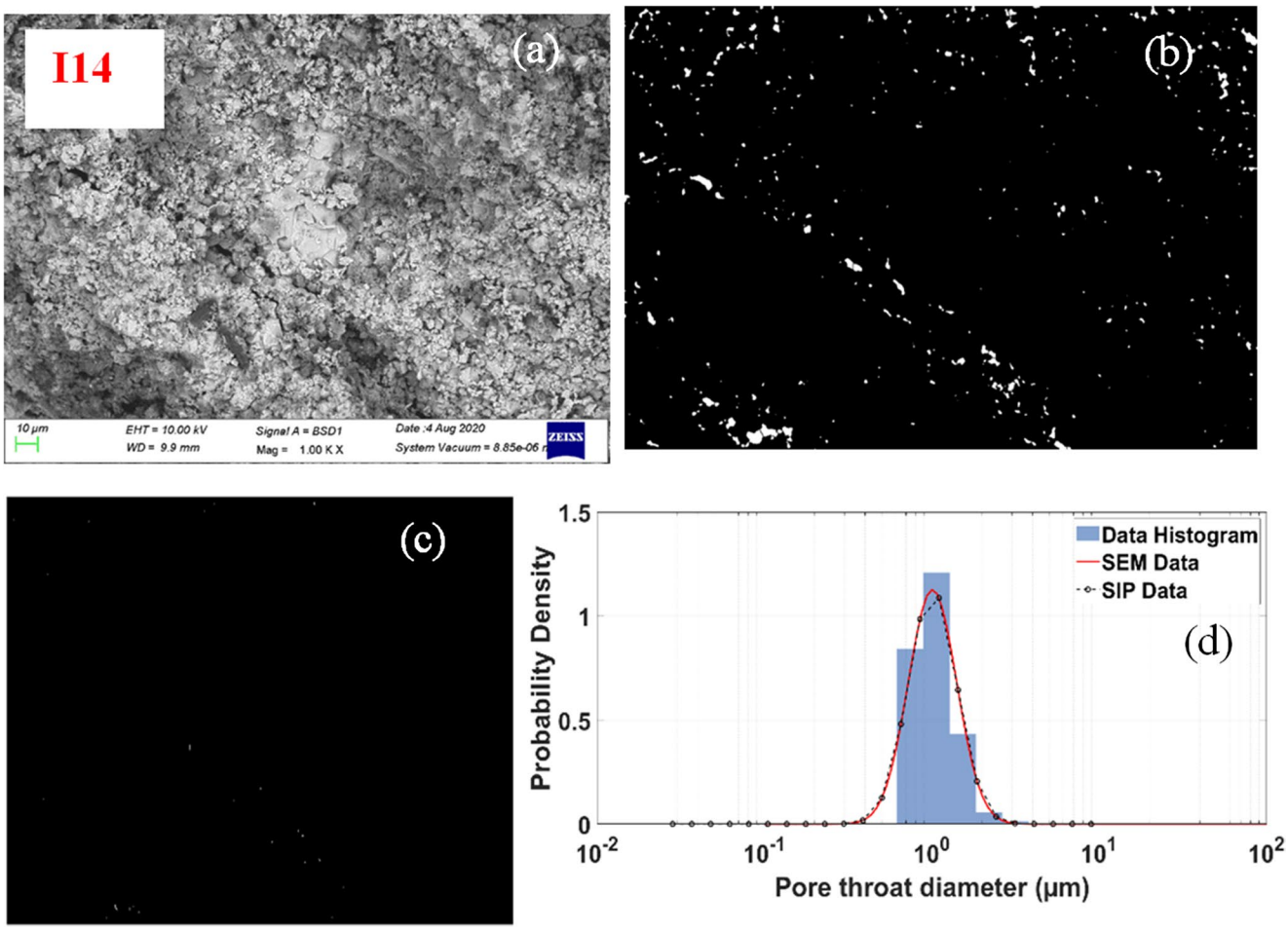


Fig. 7 (continued)

To evaluate the applicability of the Weller et al. (2015a) model to carbonate rocks, we examined the relationship between quadrature conductivity (σ'') and the product kF^2.1, where k is permeability and F is the formation factor. A clear correlation was observed for the individual dataset, and the model was fitted to the carbonate samples (Fig. 9). However, when compared alongside the modified Revil and Paris models, the Weller model showed only moderate agreement, indicating that its predictive capability is somewhat limited for this heterogeneous carbonate dataset. Additional data may be required in future work to further assess the reliability and general applicability of this model to carbonate lithologies.

To further elucidate the observed correlations between SIP-derived permeability estimates and measured values, we examined the influence of microstructural variations on electrical and flow properties. Grain size distributions.

derived from SEM image analysis indicate that wackestone samples exhibit broader distributions due to the presence of carbonate mud and larger biogenic fragments, whereas mudstone and limestone samples show more uniform grain sizes. These variations in grain geometry affect formation factor and dielectric relaxation, thereby impacting quadrature conductivity and SIP-derived permeability estimates (Fig. 10). This link highlights how microstructural heterogeneity mediates the relationship between electrical responses and petrophysical properties.

While Revil et al. (2010) emphasized the role of Stern layer polarization in the validity of their model, the close alignment of predicted and measured permeability in this study suggests that Stern layer polarization is a dominant mechanism in the analyzed carbonate rocks. This is consistent with the known mineralogical composition and textural heterogeneity of Indian offshore Tertiary carbonates (Sharma et al. 2006). Nevertheless, future work should consider incorporating mechanistic models and direct surface conductivity measurements to further validate the polarization mechanisms involved.

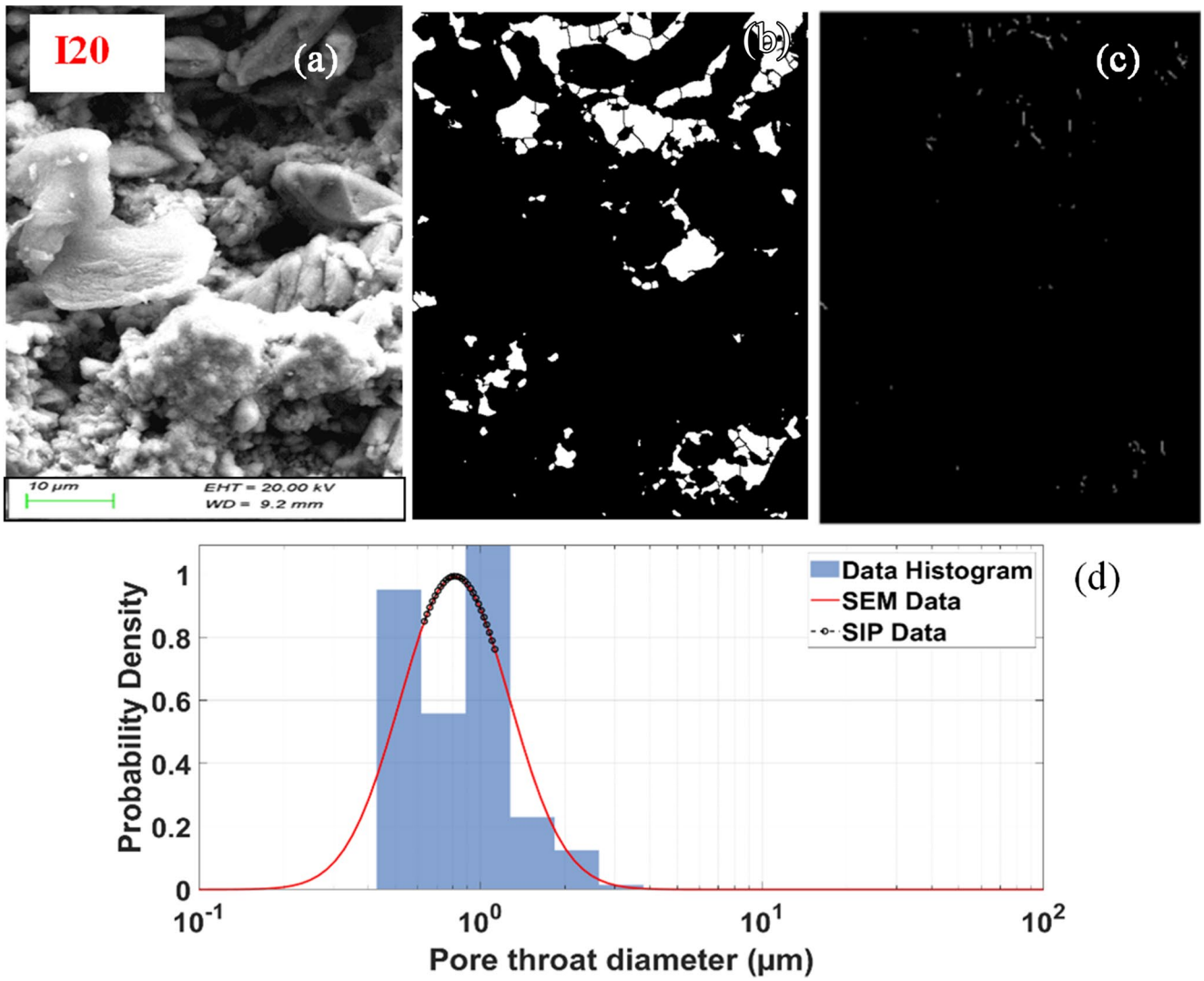


Fig. 7 (continued)

Conclusion

This study aimed to demonstrate the potential of Spectral Induced Polarization (SIP) as a non-destructive tool for estimating pore size distributions in carbonate rocks. By validating SIP-derived results against conventional techniques such as Mercury Intrusion Porosimetry (MIP) and Scanning Electron Microscopy (SEM), we established a robust experimental and analytical framework for characterizing complex carbonate pore systems.

Our analysis revealed that the commonly used Cole–Cole model fails to adequately capture the broad and multimodal relaxation behavior observed in carbonate rocks with heterogeneous textures. In contrast, the Double Cole–Cole model, which accounts for dual relaxation mechanisms, provided superior fits to the SIP spectra and yielded relaxation time distributions (RTDs) that closely aligned with pore size distributions obtained independently from MIP and SEM. The results demonstrate that grain-supported and matrix-supported porosity produce distinct relaxation domains in SIP spectra, which can be effectively resolved using the Double Cole–Cole model.

The adoption of a Double Cole–Cole formulation improved the representation of SIP spectra, particularly in samples exhibiting complex textural heterogeneity. The model theoretically allows the resolution of overlapping polarization processes that may arise from grain-supported and matrix-supported domains. While these findings conceptually strengthen the interpretation of dual relaxation behavior in carbonates, their validation through microstructural and petrophysical analyses (e.g., MIP, SEM) will be addressed in future work.

We further evaluated the applicability of SIP parameters for permeability estimation by comparing several published

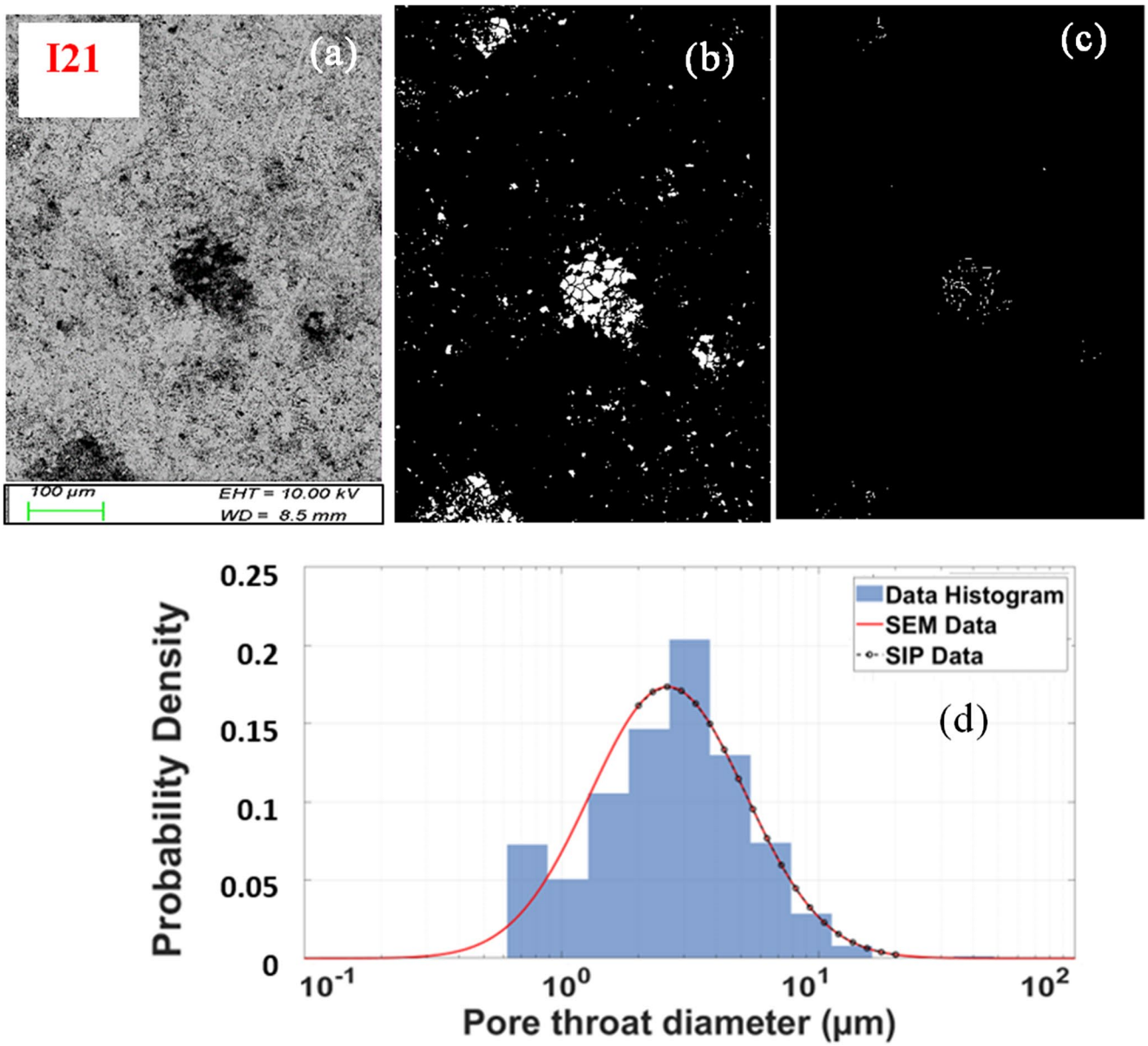


Fig. 7 (continued)

models originally developed for sandstones and modified in this study to better represent carbonate lithologies. Among them, the modified Revil and Florsch (2010) model demonstrated the best performance, showing strong agreement with measured permeability values obtained using a gas permeameter (R^2 = 0.80). The model predicts permeability based on quadrature conductivity, formation factor, and specific surface conductivity of the Stern layer, indicating that Stern layer polarization is a dominant mechanism in the analyzed carbonate rocks. Notably, the use of a constant Stern layer conductivity ($\Sigma_s = 2 \times 10^{-9}$ S) may be a limiting assumption, and future research should explore lithology-dependent variations to improve model generalizability.

Grain size distributions derived from SEM image analysis further substantiated the relationship between electrical properties and petrophysical parameters. Broader grain size distributions in wackestone samples, compared to mudstone and limestone, reflected depositional and diagenetic influences, which in turn affect formation factor and dielectric response. This underscores the dual impact of pore geometry on both fluid transport and SIP response.

The classification of pores as microporous—based on the dominant peak range (<50 µm) observed across SIP, MIP, and SEM—has critical implications for understanding carbonate reservoir performance. Microporosity influences not only storage and permeability but also saturation behavior,

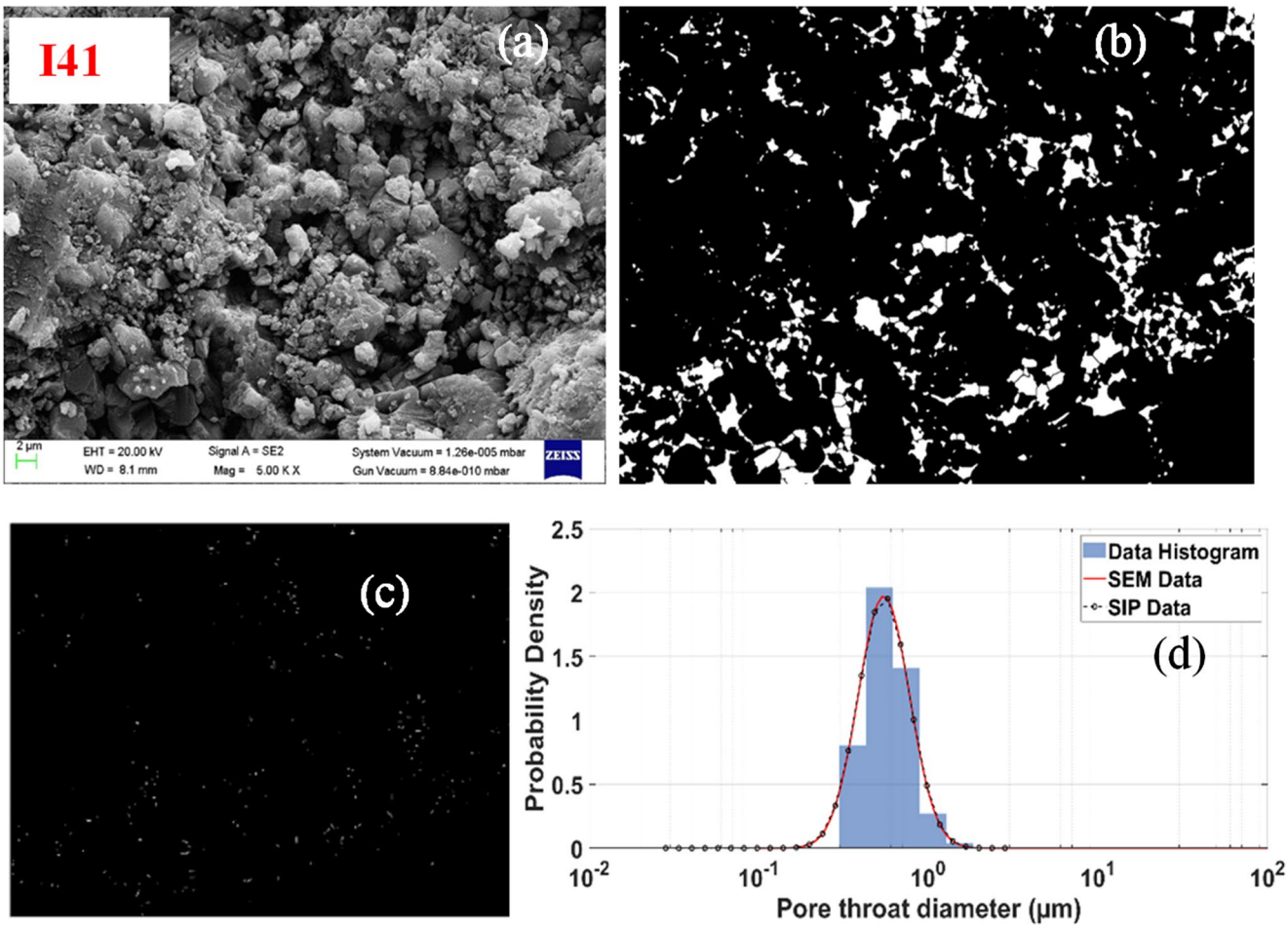


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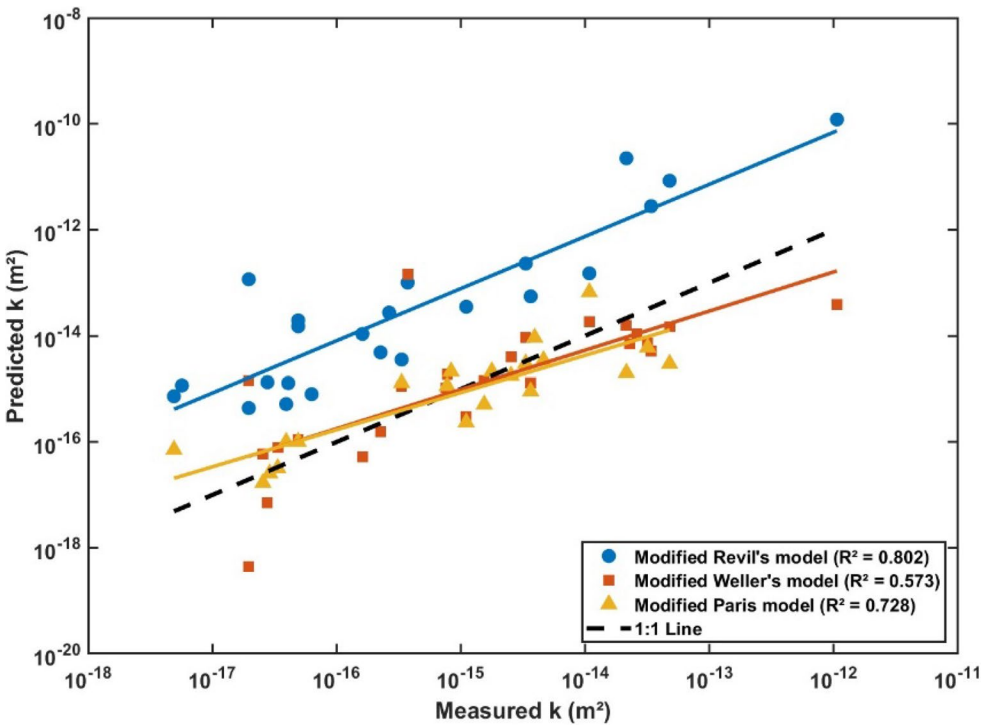


Fig. 8 Comparison between experimentally measured permeability and SIP-derived predictions using different models. The Weller et al. (2015a) formulation $K = \frac{7.14*10^{-13}}{\sigma''^{0.45}F^{2.39}}$, (square) shows moderate agreement with the measured data. The Paris model, $K = \frac{1.35*10^{-10}}{Spor^{0.82}F^{1.54}}$, (triangle), provides a reasonable correlation. While the Revil and Florsch (2010) model with m=2.1 (circle) yields a correlation of $R^2 = 0.80$. Here, σ″ represents the quadrature conductivity at 1 Hz, F is the formation factor"

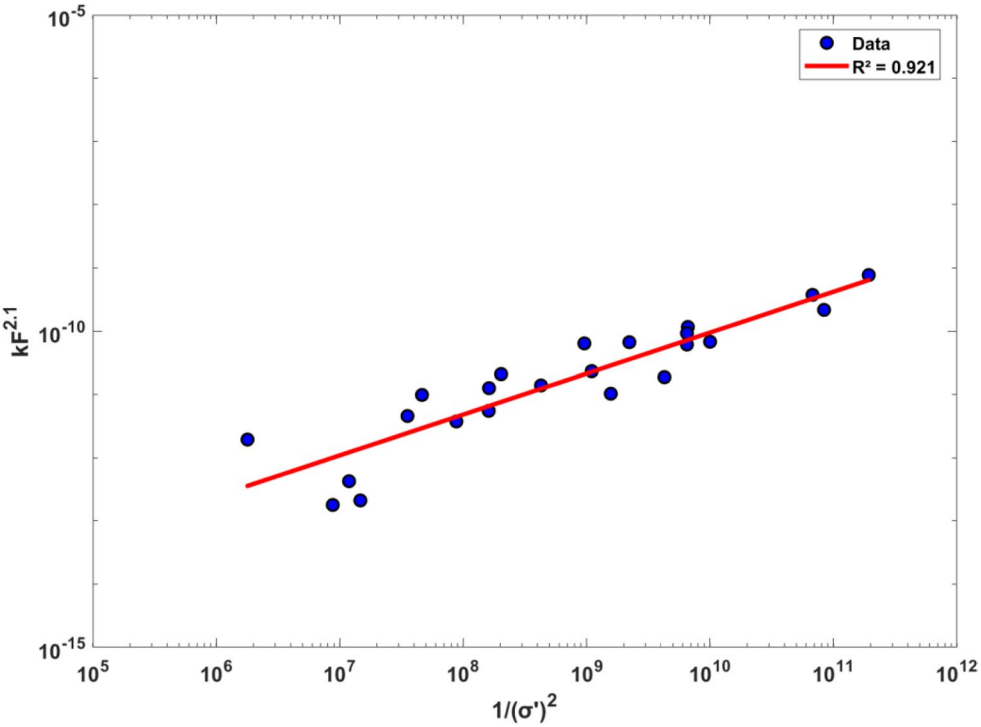


Fig. 9 Relationship between quadrature conductivity and permeability along with the formation factor. The behavior of quadrature conductivity is influenced by the fluctuations in pore structure and connectivity, revealing insights into the internal organization of grains and pores

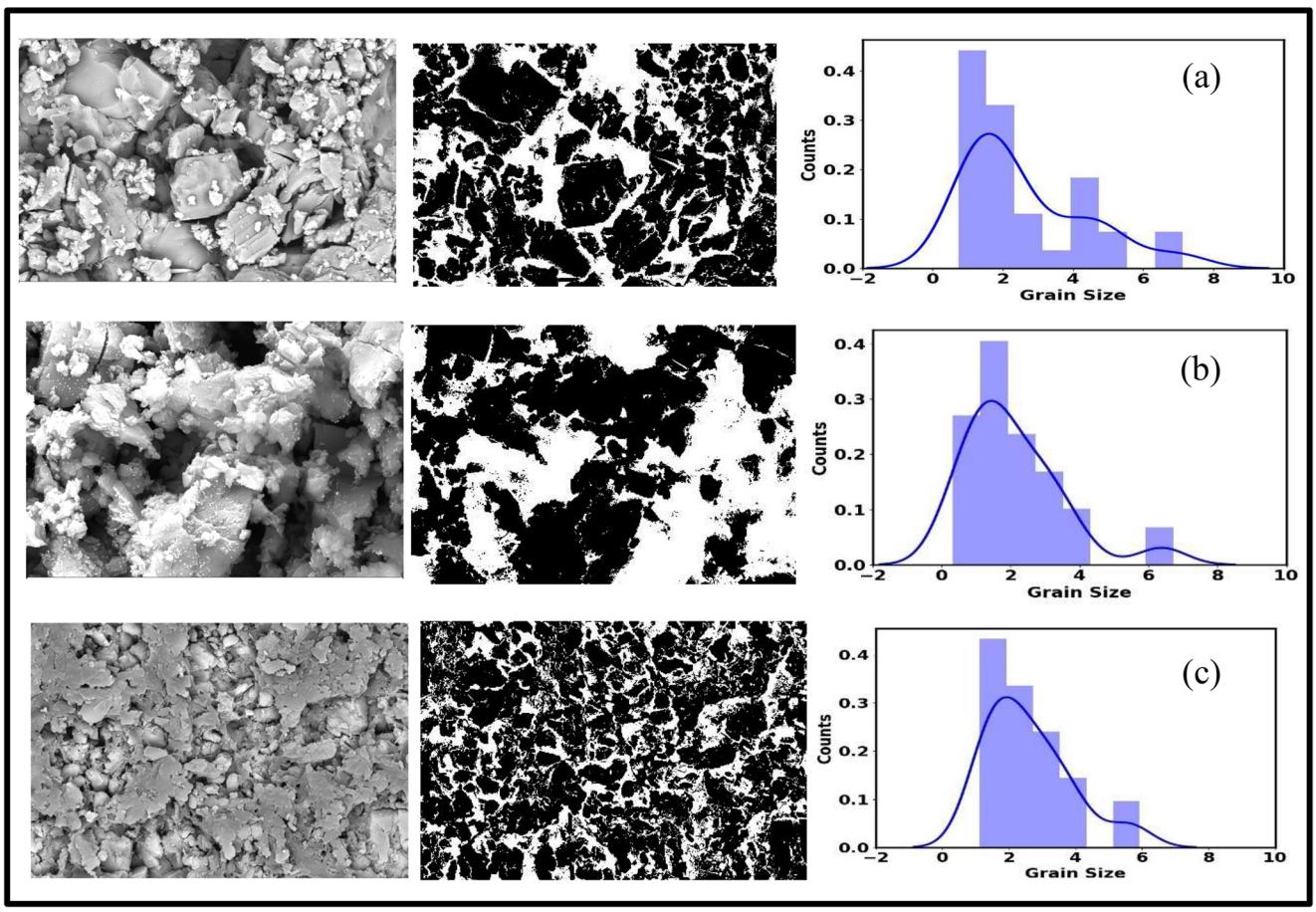


Fig. 10 The grain size distribution of (**a**) wackestone, (**b**) mudstone, and (**c**) limestone carbonate sample using SEM image analysis. The BSE SEM images are displayed on the left corner and segmented image in the middle (from left to right)

and thus accurate characterization is vital for reservoir evaluation and management.

In summary, this work bridges SIP-derived electrical measurements with microstructural and petrophysical characterization of carbonate rocks. It demonstrates that SIP, constrained and validated by MIP and SEM, can serve as a powerful, non-destructive approach for estimating pore size distribution and permeability. Future work should consider extending this framework to incorporate temperature and pressure effects and implement uncertainty quantification through Monte Carlo or Bayesian inversion to enhance reliability across varied geological settings.

Appendix A: mathematical derivation of the SIP RTD inversion framework

This appendix outlines the mathematical formulation used to derive the relaxation time distribution (RTD) from complex conductivity data. The inversion procedure is based on the generalized linear admittance model proposed by Florsch et al. (2014), adapted for improved applicability to carbonate systems.

We begin with the fundamental model for complex conductivity.

$$\sigma^* = \sigma_\infty + \frac{\sigma_0 - \sigma_\infty}{1 + i\omega\tau_0} + i\omega C \tag{A1}$$

Here, σ_0 and σ_∞ denote the low- and high-frequency limits of conductivity, respectively, τ_0 is the characteristic relaxation time, and C is a capacitive term modeling high-frequency behavior.

To extend this single-relaxation model to account for a spectrum of relaxation processes, the Dirac delta function $\delta(\tau_0 - \tau)$ is replaced by a continuous, normalized relaxation time distribution g(τ), yielding:

$$\sigma^* = a_R - M.\int_0^\infty g_\tau(\tau)\,\phi(\omega,\tau)\,d\tau + i\omega C \tag{A2}$$

where $M = (\sigma_0 - \sigma_\infty)$, and $\phi(\omega,\tau) = 1/(1 + i\omega\tau)$ serves as the kernel function linking relaxation processes to frequency-dependent response.

To improve numerical stability and accommodate scale invariance, a logarithmic transformation of variables is introduced:

$$z = -\log(\omega),\ s = \log(\tau),\ G_s(s) = \tau g(\tau),\ \varnothing(z,s) = \phi(e^{-z}, e^s)$$

Applying this transformation, the model becomes:

$$A_z(z) = a_R - \int_{-\infty}^{\infty} G_s(s)\,\varnothing(z,s)\,ds + ie^{-z}C \tag{A3}$$

This formulation allows the inversion to proceed within a time-varying generalized least squares (TVGLS) framework. The forward model is discretized using Riemann sums, and the resulting system of equations is cast into the linear form:

d = Gm.

where d is the data vector containing both real and imaginary parts of the measured conductivity spectrum, G is the sensitivity matrix constructed from the kernel functions, and m represents the unknown RTD.

Overlap with Florsch et al. (2014): The formulation up to this stage follows the generalized linear admittance model of Florsch et al. (2014), including the linearization and TVGLS inversion framework.

Novel contributions in the present work:

1. Double Cole–Cole initialization: The inversion is initialized using parameters derived from a Double Cole–Cole model to capture bimodal or overlapping polarization processes commonly observed in heterogeneous carbonates:

$$\sigma^*_\omega = \sigma_\infty\left[1 - \frac{M_1}{1 + i\omega\tau_1{}^{c1}} - \frac{M_2}{1 + i\omega\tau_2{}^{c2}}\right] \tag{A4}$$

The parameters M_1 and M_2 (chargeabilities), τ_1 and τ_2 (relaxation times), and c_1 and c_2 (Cole-Cole exponents) are determined through nonlinear curve-fitting of the measured complex conductivity spectra.

2. Carbonate-specific adaptations: This framework explicitly considers heterogeneous pore structures in carbonate rocks, enabling accurate RTD estimation in lithologies exhibiting overlapping relaxation times.

The real and imaginary components of the Double Cole–Cole model are:

$$\sigma' = \sigma_\infty\left[1 - \frac{M_1\omega\tau_1{}^{c1}\left(c_1\cos\frac{\pi}{2}\right)}{(1+\omega\tau_1{}^{c1}(c_1\cos\frac{\pi}{2})^2 + (\omega\tau_1{}^{c1}\sin\frac{\pi}{2}c_1)^2} - \frac{M_2\omega\tau_2{}^{c2}\left(c_2\cos\frac{\pi}{2}\right)}{(1+\omega\tau_2{}^{c2}(c_2\cos\frac{\pi}{2})^2 + (\omega\tau_2{}^{c2}\sin\frac{\pi}{2}c_2)^2}\right] \tag{A5}$$

$$\sigma'' = \sigma_\infty\left[\frac{M_1\omega\tau_1{}^{c1}\left(c_1\sin\frac{\pi}{2}\right)}{(1+\omega\tau_1{}^{c1}(c_1\cos\frac{\pi}{2})^2 + (\omega\tau_1{}^{c1}\sin\frac{\pi}{2}c_1)^2} - \frac{M_2\omega\tau_2{}^{c2}\left(c_2\sin\frac{\pi}{2}\right)}{(1+\omega\tau_2{}^{c2}(c_2\cos\frac{\pi}{2})^2 + (\omega\tau_2{}^{c2}\sin\frac{\pi}{2}c_2)^2}\right] \tag{A6}$$

Finally, the quality of the RTD inversion is evaluated using the root mean square error (RMSE) (Maineult et. al., 2018):

$$RMSE = \sqrt{\frac{1}{N}\sum\left[\log_{10}\left(\sigma^*_{predicted}\right) - \log_{10}\left(\sigma^*_{observed}\right)\right]^2}$$

In conclusion, the present methodology extends the work of Florsch et al. (2014) by incorporating a Double Cole–Cole model for heterogeneous carbonate systems and by

explicitly accounting for bimodal relaxation processes, thereby enabling robust extraction of pore-scale features from SIP measurements in complex lithologies.

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Author contributions N. Panwar: Methodology, Investigation, Writing-original draftR. Sharma: Methodology, Investigation, Conceptualization, Supervision, Writing-original draftH.J. Kalita: Writing- review & editingA. Dhiman: Coding- editing, Writing- review & editingP. Soni: Coding- editing, Writing- review & editingS. Malik: Writing-review & editingP. Pandey: Writing- review & editing.

Data availability Data and code would be shared if a request is received for the same.

Declarations

Competing interests The authors declare no competing interests.

References

Abell AB, Willis KL, Lange DA (1999) Mercury intrusion porosimetry and image analysis of cement-based materials. J Colloid Interface Sci 211:39–44

August H, Azizoglu Z, Heidari Z, Goncalves L, Oliveira LA, Neto MS, Victor RA (2025) Novel Estimation of pore-throat size and permeability in carbonates from the integration of NMR and resistivity data in the Santos basin of Brazil. Pet Geosci. https://doi.org/10.1144/petgeo2024-093

B¨orner FD, Sch¨on JH (1991) A relation between the quadrature component of electrical conductivity and the specific surface area of sedimentary rocks. Log Anal 32:612–613

Bohnsack D, Potten M, Freitag S, Einsiedl F, Zosseder K (2021) Stress sensitivity of porosity and permeability under varying hydrostatic stress conditions for different carbonate rock types of the geothermal Malm reservoir in Southern Germany. Geotherm Energy. https://doi.org/10.1186/s40517-021-00197-w

Brown E (2011) Successfully stimulating complex carbonate reservoirs. Society of Petroleum Engineers

Bust VK, Oletu JU, Worthington PF (2011) The challenges for carbonate petrophysics in petroleum resource Estimation. Society of Petroleum Engineers

Cao Z, Liu G, Zhan H, Li C, You Y, Yang C, Jiang H (2016) Pore structure characterization of Chang-7 tight sandstone using MICP combined with N2GA techniques and its geological control factors. Sci Rep. https://doi.org/10.1038/srep36919

Choquette PW, Pray LC (1970) Geologic Nomenclature and Classification of Porosity in Sedimentary Carbonates. Am Assoc Petroleum Geol Bull 54:207–250

Cnudde V, Cwirzen A, Masschaele B, Jacobs PJ (2009) Porosity and microstructure characterization of Building stones and concretes. Eng Geol. https://doi.org/10.1016/j.enggeo.2008.06.014

Coates GR, Xiao L, Prammer MG (1999) NMR logging: principles & applications. Halliburton Energy Services

Coperey A, Revil A, Abdulsamad F, Stutz B, Duvillard PA, Ravanel L (2019) Low-frequency induced polarization of porous media undergoing freezing: preliminary observations and modeling. JGR Solid Earth. https://doi.org/10.1029/2018JB017015

Cunningham KJ, Sukop M, Curran HA (2012) Carbonate Aquifers. In *Developments in Sedimentology* (pp. 869–896). https://doi.org/10.1016/B978-0-444-53813-0.00028-9

Diamond S, Leeman ME (1994) Pore size distributions in hardened cement paste by Sem image analysis. MRS Online Proc Libr 370:217–226. https://doi.org/10.1557/PROC-370-217

Florsch N, Revil A, Camerlynck C (2014) Inversion of generalized relaxation time distributions with optimized damping parameter. J Appl Geophys 109. https://doi.org/10.1016/j.jappgeo.2014.07.013

Garcia AP, Han Y, Heidari Z (2018) Integrated workflow to estimate permeability through quantification of rock fabric using joint interpretation of nuclear magnetic resonance and electric measurements. Petrophysics 59:672–693. https://doi.org/10.30632/PJV59N5-2018a7

Garcia A, Smith EA, Glaser DR, Slater L (2024) Spectral induced polarization responses of carbonate precipitation in gel-saturated media with and without electron conductors. Geophys J Int 34–48. https://doi.org/10.1093/gji/ggad411

Gomaa MM, Kassab MA, El-Sayed NA (2015) Study of pertrographical and electrical properties of some jurassic carbonate rocks, North Sinai, Egypt. Egypt J Pet. https://doi.org/10.1016/j.ejpe.2015.07.003

Gonzalez RC, Woods RE (2018) Digital Image Processing, Fourth Edition. Pearson Education Limited, 786–793, ISBN 978-0-13-335672-4

Halisch M, Weller A, Hupfer S, Kassab MA (2014) Impedance Spectroscopy on Carbonates. *Annual Symposium of the Society of Core Analysts*. Avignon, France. https://doi.org/10.13140/2.1.3079.9045

Hussein D, Lawrence J, Rashid F, Glover P, Lorinczi P (2018) Engineering in chalk. London-UK. https://doi.org/10.1680/eiccf.64072.529

Irajian AA, Guilani KB, Mahari R, Solgi A (2016) Porosity and Rock-Typing in hydrocarbon reservoirs case study in upper member of Dalan formation in Kish gas Field, South of Zagros, Iran. Open J Geol 399–409. https://doi.org/10.4236/ojg.2016.66035

Jougnot D, Ghorbani A, Revil A, Leroy P, Cosenza P (2010) Spectral induced polarization of partially saturated clay-rocks: a mechanistic approach. Geophys J Int. https://doi.org/10.1111/j.1365-246X.2009.04426.x

Kargarpour MA (2020) Carbonate reservoir characterization: an integrated approach. J Pet Explor Prod Technol 10. https://doi.org/10.1007/s13202-020-00946-w

Kovács A (2021) Quantitative classification of carbonate aquifers based on hydrodynamic behaviour. Hydrogeol J Volume 29:33–52

Li C, Chen G, Qiu X, Lou Q, Gao X (2021) A direct proof for Maxwell–Wagner effect of heterogeneous interface. AIP Adv. https://doi.org/10.1063/5.0040947

Liu X, Kong L, Zhou K (2014) A time domain induced polarization relaxation time spectrum inversion method based on a damping factor and residual correction. Pet Sci 11:519–525. https://doi.org/10.1007/s12182-014-0367-9

Liu J, Song S, Cao X, Meng Q, Pu H, Wang Y, Liu J (2020) Determination of full-scale pore size distribution of Gaomiaozi bentonite. J Rock Mech Geotech Eng 403–413. https://doi.org/10.1016/j.jrmge.2019.12.005

Lonoy A (2006a) Making sense of carbonate pore systems. AAPG Bull 90(9):1381–1405

Lonoy A (2006b) Making sense of carbonate pore systems: American association of petroleum geologists (AAPG) bulletin. 90:1381–1405. https://doi.org/10.1306/03130605104

Lu H, Tang H, Wang M, Li X, Zhang L, Wang Q, Zhao Y, Zhao Y, Liao J (2021) Pore structure characteristics and permeability prediction model in a cretaceous carbonate reservoir, North Persian Gulf basin. Geofluids. https://doi.org/10.1155/2021/8876679

Lucia FJ (1983) Petrophysical parameters estimated from visual descriptions of carbonate rocks: a field classification of carbonate pore space. J Pet Technol. https://doi.org/10.2118/10073-PA

Lucia FJ Rock-Fabric/Petrophysical Classification of Carbonate Pore Space for Reservoir Characterization. AAPG Bulletin, 1275–1300. doi:, Lucia FJ (1995) (2007). Carbonate Reservoir Characterization. Berlin, Heidelberg: Springer. doi:https://doi.org/10.1007/978-3-540-72742-2

Maineult A, Jougnot D, Revil A (2018) Variations of petrophysical properties and spectral induced polarization in response to drainage and imbibition: a study on a correlated random tube network. Geophys J Int 212(2):1398–1411. https://doi.org/10.1093/gji/ggx474

Markov M, Kazatchenko E, Mousatov A, Pervago E (2012) The dielectric permittivity of carbonate formations from the unified microstructure model. J Appl Geophys 76:56–63

Muller-Huber E, B¨orner F, B¨orner JH, Kulke D (2018) Combined interpretation of NMR, MICP, and SIP measurements on mud-dominated and grain-dominated carbonate rocks. J Appl Geophys 159:228–240

Niu Q, Zhang C (2018) Joint inversion of NMR and SIP data to estimate pore size distribution of geomaterials. Geophys J Int. https://doi.org/10.1093/gji/ggx501

Osterman G, Keating K, Binley A, Slater L (2016) A laboratory study to estimate pore geometric parameters of sandstones using complex conductivity and nuclear magnetic resonance for permeability prediction. Water Resour Res 52:4321–4337. https://doi.org/10.1002/2015WR018472

Panwar N, Revil A, Sharma R, Schmutz M, Duvillard PA, Garcia B, Cerepi A, Garcia-Artigas R, Vaudelet P, Malet E, Jaillet S (2021) Induced polarization of carbonates. Journal of Geophysical Research: Solid Earth 126(6):e2021JB022029

Pape H, Riepe L, Schopper J (1982) A Pigeon-hole model for relating permeability to specific surface. The Log Analyst. Society of Petrophysicists and Well Log Analysts 23(1). Errata

Pape H, Riepe L, Schopper JR (1987) Theory of self-similar network structures in sedimentary and igneous rocks and their investigation with microscopical and physical methods. J Microsc 148(2):121–47

Pelton WH, Ward SH, Hallof PG, Sill WR, Nelson PH (1978) Mineral discrimination and removal of inductive coupling with multifrequency IP. Geophysics. https://doi.org/10.1190/1.1440839

Revil A, Florsch N (2010) Determination of permeability from spectral induced polarization in granular media. Geophys J Int 1480–1498. https://doi.org/10.1111/j.1365-246X.2010.04573.x

Revil A, Koch K, Holliger K (2012) Is it the grain size or the characteristic pore size that controls the induced polarization relaxation time of clean sands and sandstones? Water Resour Res W05602. https://doi.org/10.1029/2011WR011561

Revil A, Florsch N, Camerlynck C (2014) Spectral induced polarization porosimetry. Geophys J Int. https://doi.org/10.1093/gji/ggu180

Revil A, Coperey A, Shao Z, Florsch N, Fabricius IL, Deng Y, Delsman JR, Pauw PS, Karaoulis M, Louw PGB de., van Baaren ES, Dabekaussen W, Menkovic A, Gunnink JL (2017) Complex conductivity of soils. Water Resour Res 7121–7147. https://doi.org/10.1002/2017WR020655

Safari H, Balcom BJ, Afrough A (2021a) Characterization of pore and grain size distributions in porous geological samples- an image processing workflow. Comput Geosci 156:104895. https://doi.org/10.1016/j.cageo.2021.104895

Safari H, Balcom BJ, Afrough A (2021b) Characterization of pore and grain size distributions in porous geological samples – an image processing workflow. Comput Geosci 104895. https://doi.org/10.1016/j.cageo.2021.104895

Salvini S, Coletti C, Maritan L et al (2023) Exploring the pore system of carbonate rocks through a multi-analytical approach. Environ Earth Sci 82:564. https://doi.org/10.1007/s12665-023-11234-1

Schindelin J, Arganda-Carreras I, Frise E, Kaynig V, Longair M, Pietzsch T, Cardona A (2012) Fiji: an open-source platform for biological-image analysis. Nat Methods 9(7):676–682. https://doi.org/10.1038/nmeth.2019

Schneider CA, Rasband WS, Eliceiri KW (2012) NIH image to imageJ: 25 years of image analysis. Nat Methods 9(7):671–675. https://doi.org/10.1038/nmeth.2089

Schwartz N, Furman A (2012) Spectral induced polarization signature of soil contaminated by organic pollutant: experiment and modeling. Journal of Geophysical Research: Solid Earth. https://doi.org/10.1029/2012JB009543

Schwarz G (1962) A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution. J Phys Chem 66:2636–2642. https://doi.org/10.1021/j100818a067

Sharma R, Prasad M, Surve G, Katiyar GC (2006) October. On the applicability of Gassmann model in carbonates. In SEG International Exposition and Annual Meeting (pp. SEG-2006). SEG

Sharma R, Prasad M, Batzle M, Vega S (2013) Sensitivity of flow and elastic properties to fabric heterogeneity in carbonates. Geophys Prospect. https://doi.org/10.1111/1365-2478.12030

Stingaciu LR, Weihermüller L, Pohlmeier SH, Stapf S, Vereecken H, Pohlmeier A (2010) Determination of pore size distribution and hydraulic properties using nuclear magnetic resonance relaxometry: A comparative study of laboratory methods. Water Resour Res W11510. https://doi.org/10.1029/2009WR008686

Sun B, Yang Q, Zhu J, Shao T, Yang Y, Hou C, Li G (2020) Pore size distributions and pore multifractal characteristics of medium and low rank coals. Sci Rep. https://doi.org/10.1038/s41598-020-79338-3

Tarantola A, Valette B (1982) Generalized nonlinear inverse problems solved using the least squares criterion. Rev Geophys Space Phys. https://doi.org/10.1029/RG020i002p00219

Tartrat T, Revil A, Abdulsamad F, Ghorbani A, Jougnot D, Coperey A, Vaissiere Rd (2019) Induced polarization response of porous media with metallic particles- part 10: influence of desiccation. Geophysics E357–E375. https://doi.org/10.1190/GEO2019-0048.1

Tong M, Li L, Wang W, Jiang Y (2006) Determining capillary-pressure curve, pore-size distribution, and permeability from induced polarization of Shaley sand. Geophysics N33:N40. https://doi.org/10.1190/1.2195989

Vinegar HJ, Waxman MH (1984) Induced polrization of shaly sands. Geophysics 49:1267–1287

Vinegar HJ, Waxman MH (1988) In situ induced polarization method for determining formation permeability:United States Patent No 4:743854

Washburn EW (1921) Note on a method of determining the distribution of pore sizes in a porous material. Proc Natl Acad Sci U S A. https://doi.org/10.1073/pnas.7.4.115

Weiqiang L, Longxin M, Lun Z, Jianxin L, Shuqin W, Zifei F, Dali S, Changhai L, Fachao S, Wenqi Z, Meng S (2020) Pore-throat structure characteristics and its impact on the porosity and permeability relationship of carboniferous carbonate reservoirs in Eastern edge of Pre-Caspian basin. Petroleum Explor and Dev 47 1027–1041

Weller A, Slater L, Nordsiek S, Ntarlagiannis D (2010b) On the estimation of specific surface per unit pore volume from induced polarization: a robust empirical relation fits multiple data sets. Geophysics 75:WA105–WA112

Weller A, Slater L, Binley A, Nordsiek S, Xu S (2015a) Permeability prediction based on induced polarization: insights from measurements on sandstone and unconsolidated samples spanning a wide permeability range. Geophysics 80:D161–D173

Widiatmoko E, Abdullah M, Khairurrijal (2010) A Method to Measure Pore Size Distribution of Porous Materials Using Scanning Electron Microscopy Images. AIP Conference Proceedings, (p. 1284). https://doi.org/10.1063/1.3515554

Yao J, Hu R, Wang C, Yang Y (2014) Multiscale pore structure analysis in carbonate rocks. Int J Multiscale Comput Eng. https://doi.org/10.1615/IntJMultcompEng.2014010500

Zhang Y, Jin S, Wang Y, Wang Y (2015) Characterization of the pore size distribution with SEM images processing for the tight rock, IEEE International Conference on Information and Automation, Lijiang, China, 2015, pp. 653–656. https://doi.org/10.1109/ICInfA.2015.7279367

Zhang Z, Kruschwitz S, Weller A, Halisch M (2018a) Solid Earth, Enhanced pore space analysis by use of μ-CT, NMR, MIP and SIP. 1225–1238. https://doi.org/10.5194/se-9-1225-2018

Zhang F, Niu Q, Zhang C (2018b) Estimating pore size distribution in carbonate reservoir rocks using joint inversion of NMR and complex conductivity data: SEG International Exposition and 88th Annual Meeting

Zhang T, Li Z, Lai F, Adenutsi CD, You Q (2022) Effects of the sandstone pore structure on spontaneous imbibition: a systematic experimental investigation based on fractal analysis. Energy Fuels 382(396). https://doi.org/10.1021/acs.energyfuels.1c03696

Zheng S, Yao Y, Elsworth D, Wang B, Liu Y (2020) A novel pore size classification method of coals: investigation based on NMR relaxation. J Nat Gas Sci Eng. https://doi.org/10.1016/j.jngse.2020.103466

Zimmermann E, Kemna A, Berwix J, Glaas W, Munch HM, Huisman JA (2008) A high-accuracy impedance spectrometer for measuring sediments with low polarizability. Meas Sci Technol 19

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